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TOPICAL REVIEW

Design and Deployment of Swarm Engineering Systems: Insights, Challenges, and Innovations

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ABSTRACT In this study, we explore the fundamentals of design and deployment of Multi-Agent Swarm Engineering Systems (MASESs), a comprehensive framework aimed at improving decision-making and operational efficiency in industrial environments. Swarm engineering systems draw inspiration from collective behaviors observed in natural swarms (such as ants or bees), offering a decentralized approach to problem-solving in complex systems. We provide a detailed theoretical foundation for MASES, focusing on key elements such as spatial control formation, adaptive coordination, decision-making dynamics, and critical factors such as scalability, adaptability, and robustness in practical applications. The analysis is enriched with insightful tables that highlight current innovations and challenges, underscoring the importance of balancing decentralization with scalability. In transitioning to practical applications, we discuss design methodologies and performance evaluation strategies, emphasizing the transformative potential of swarm engineering systems. Crucially, we highlight how artificial intelligence (AI) and machine learning (ML) are being integrated into MASES to enhance decision-making capabilities, adaptability, and scalability. Advanced frameworks and simulation environments have been shown to play a crucial role in ensuring that MASES operates effectively, fostering intelligence and adaptability to meet complex industrial demands. By showcasing various applications, we highlight the potential of MASES to enhance operational landscapes across industries and identify pathways for future research to refine and perfect this integration.

INDEX TERMS Swarm intelligence, decentralized systems, multi-agent systems, design approaches, development frameworks.

I. FRAME OF REFERENCE

Multi-agent swarm engineering systems (MASESs) draw inspiration from natural phenomena such as ant foraging, bird flocking, and fish schooling, applying the principles of swarm intelligence to solve complex, real-world challenges. These systems rely on decentralized, self-organizing behavior, where autonomous agents interact locally to achieve collective goals without centralized control [1]. This approach is particularly advantageous in dynamic and uncertain environments that demand flexibility, robustness, and adaptability.

As MASES become more prevalent, understanding ethical implications such as accountability, transparency, and data privacy, along with fostering effective human-swarm interaction to ensure appropriate human oversight and build trust in autonomous operations, becomes paramount for responsible deployment. MASES leverages decentralized and distributed control to enable autonomous collaboration among agents across diverse domains. Real-world applications demonstrate the system's versatility in areas such as warehouse automation, precision agriculture, environmental monitoring, and

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medical robotics [2], [3]. These applications directly benefit from the following foundational design principles:

- Decentralization supports autonomous robot operations in logistics, allowing robots to perform tasks such as inventory scanning and restocking independently [4].
- Scalability enables the deployment of large drone fleets for agricultural monitoring and environmental data collection [5].
- Adaptability: MASES can respond effectively to changes in environmental conditions or task requirements, ensuring consistent performance in dynamic and unpredictable settings [6].
- Robustness ensures fault tolerance and reliability through self-healing mechanisms, distributed control, and advanced error-handling tools [7].

This extends to resilience, enabling MASES to maintain function and recover effectively even when facing significant agent failures, unexpected environmental disturbances, or communication disruptions. Mechanisms like redundancy, adaptive re-planning, and decentralized decision-making contribute significantly to this enhanced robustness.

Together, these principles establish the core of MASES, enabling smart, scalable, and resilient industrial ecosystems. By integrating cutting-edge technologies and systematic design practices, MASES bridges theoretical insights with practical implementations. Building on these principles, MASES has become a powerful paradigm for addressing distributed, large-scale problems in fields such as robotics, telecommunications, process optimization, and intelligent scheduling [8]. These systems capitalize on self-organization to respond to environmental uncertainty and emergent task complexity.

Despite their potential, swarm systems face significant design challenges, such as achieving distributed consensus, maintaining scalability, and ensuring fault tolerance [9]. Beyond these, real-world deployment introduces further complexities, notably system interoperability across heterogeneous platforms, robust sensor fusion from diverse data streams, and seamless data integration from various environmental and operational sources. Structured design methodologies and simulation platforms are essential to overcoming these barriers. Tools such as Gazebo and the Robot Operating System (ROS) facilitate iterative testing and deployment. Moreover, AI-based decentralized algorithms, such as Bayes Swarm, support adaptive coordination via asynchronous operations and dynamic rebalancing, improving performance in uncertain environments [10], [11]. This integration of AI and ML is pivotal for enabling agents to learn optimal strategies, predict environmental changes, and adapt collective behaviors, thereby directly enhancing the scalability and responsiveness of MASES in dynamic scenarios.

Recent advancements further highlight the power of bio-inspired and AI-based optimization methods in enhancing system performance, as seen in studies like 'Enhanced MPPT Efficiency in Photovoltaic Systems with a New Artificial Protozoa Optimizer Method,' 'Enhancing photovoltaic performance using artificial intelligence methods based on Fuzzy Logic,' and 'A comprehensive approach to wind turbine optimization integrating MPPT and speed control.' These works underscore the growing synergy between swarm principles and advanced AI for solving complex optimization challenges in diverse engineering domains, including energy management, which directly informs the broader applicability of MASES [12]. However, achieving this requires a deep understanding of emergent behaviors and the development of reliable coordination strategies. These behaviors often arise from simple local rules yet yield intricate global patterns [13]. The self-organizing capabilities of MASES allow them to adapt in real time, making them suitable for mission-critical tasks. Designing such systems demands clear objectives and systematic processes for testing and validation. Simulation tools are indispensable for evaluating interactions and emergent behaviors in controlled settings. Nevertheless, real-world testing remains critical to assess performance under unpredictable conditions. Bridging simulation with empirical validation ensures scalability and applicability across real-world domains [14].

While existing reviews have provided valuable insights into the theoretical underpinnings of swarm intelligence [15] and the algorithmic aspects of multi-agent coordination [16], this study distinguishes itself by adopting a comprehensive engineering perspective on MASES. Our study focuses on the practical design methodologies, development frameworks, and critical challenges associated with translating theoretical swarm principles into robust and deployable engineering systems for real-world industrial applications. Unlike reviews primarily centered on specific algorithms or theoretical models, we provide a holistic view encompassing core design principles, simulation-based development strategies with an emphasis on empirical validation, and a categorization of design approaches based on their suitability for achieving key engineering metrics such as scalability, adaptability, and robustness. Furthermore, our in-depth summaries of researcher contributions presented in Tables 4, 7, and 13 offer a unique synthesis of practical advancements and remaining gaps in the field, directly addressing the challenges hindering widespread real-world adoption.

In this study, we adopted a comprehensive approach, combining theoretical exploration with practical perspectives. We conducted a structured literature review using leading academic sources such as Google Scholar, IEEE Xplore, and SpringerLink. Search terms included "swarm engineering," "multi-agent systems," "decentralized control," and "autonomous systems." This strategy ensured the inclusion of both foundational theories and the latest advancements in MASES research. We emphasize simulation-based development while acknowledging the essential role of empirical validation. The models and tools discussed are selected for their potential to transition effectively from theoretical constructs to real-world deployment. The key contributions of this paper include:

1. Exploration of Core Design Principles:

In this section, we introduce the foundational principles of MASES, including spatial control formation, adaptive coordination, decision-making dynamics, and human-swarm interaction. Core design metrics essential for the effective operation of decentralized and autonomous systems are highlighted. These metrics are implemented through approaches such as rule-based design, computational optimization, and reinforcement learning, which allow MASES to function effectively in dynamic and unpredictable environments.

2. Development Frameworks and Simulation Tools:

Frameworks such as simulation suites and cognitive agent-based environments are explored as essential components for designing, simulating, and validating MASES applications in fields such as robotics, agriculture, healthcare, transportation, and defense. This review emphasizes the importance of these frameworks in supporting real-world validation, ensuring scalability, and enhancing adaptability, enabling MASES to perform effectively in complex and rapidly changing environments.

3. Insightful Contributions via Summaries:

Comprehensive tables summarize key progress in addressing MASES design metrics, highlighting major contributions by researchers. These summaries provide valuable information for shaping future research agendas and guiding the study and deployment of MASES.

In this study, we aim to address current challenges by offering actionable recommendations to advance the research and deployment of MASES. These challenges include improving system robustness, refining development tools, and bridging the gap between theoretical advancements and practical applications in real-world settings.

Section I: This section introduces MASES as a multidisciplinary field, outlining its key concepts and the challenges associated with its design and development. It highlights the importance of swarm-based solutions for tackling complex problems and provides a comprehensive overview of the MASES architecture. Additionally, it summarizes the primary contributions of this study, establishing the context for the following sections.

Section II: This section explores the fundamental principles of MASES, including spatial control formation, adaptive swarm coordination, decision-making dynamics, and human-swarm collaboration. It also elaborates on the core design metrics that are critical for the effective implementation of decentralized and autonomous systems.

Section III: This section focuses on the methodologies for designing MASES, emphasizing the translation of theoretical principles into practical strategies. Key areas include system modeling and performance evaluation, which are essential for ensuring the effectiveness and reliability of MASES.

Section IV: This section discusses the development frameworks and tools that support MASES implementation, such as simulation platforms, cognitive agent-based frameworks, and simulation environments tailored to real-world applications. These tools play a vital role in validating scalability, adaptability, and robustness in MASES.

Section V: This section examines the real-world applications of MASES in fields such as robotics and agriculture. It reviews the contributions of MASES to these domains and provides insights into future directions for research and practical deployment.

Section VI: This section analyzes the challenges, opportunities, and effectiveness of various MASES approaches. It presents critical insights into the development and application of MASES, focusing on strategies for overcoming current limitations and maximizing their potential.

Section VII: The conclusion summarizes the key findings and contributions of the paper, emphasizing the importance of addressing core design metrics, such as decentralization, scalability, adaptability, and robustness, for the successful development and deployment of MASES.

Overall, in the study, we provide a comprehensive survey of multi-agent swarm-engineering systems, bridging the gap between theoretical foundations and practical applications. It highlights the transformative potential of MASES in addressing complex challenges across diverse industries and underscores the importance of robust design principles to ensure their success. Building upon this foundational understanding and the contributions outlined, the subsequent sections will delve deeper into the specific fundamentals of swarm engineering, detailed design analysis, and the essential development processes and tools that enable robust MASES design and deployment.

II. FUNDAMENTALS OF SWARM ENGINEERING

Multi-agent swarm-engineering systems (MASES) utilize decentralized, self-organizing systems to develop scalable, adaptive, and robust solutions for addressing complex real-world challenges [17]. By leveraging core principles such as decentralization, self-organization, and emergent behavior, MASES provides efficient and adaptive solutions for intricate tasks across diverse domains, including robotics, healthcare, and agriculture. These principles are the foundation of MASES, enabling it to operate effectively in dynamic and decentralized environments. The spatial control formation section explores mechanisms that allow agents to dynamically adjust their relative positions, forming coordinated structures critical for scalable, adaptive, and robust systems [18]. The subsequent section on adaptive swarm coordination delves into how swarm systems modify their behaviors and configurations in response to shifting environmental conditions and task requirements. The decision-making dynamics section highlights how agents, through both individual and collective decisions, achieve coordination and efficiency in swarm behaviors [19]. Finally, the human-swarm collaboration section focuses on interfaces between human operators and swarm systems, emphasizing the pivotal role of human interaction in enhancing system resilience and functionality.

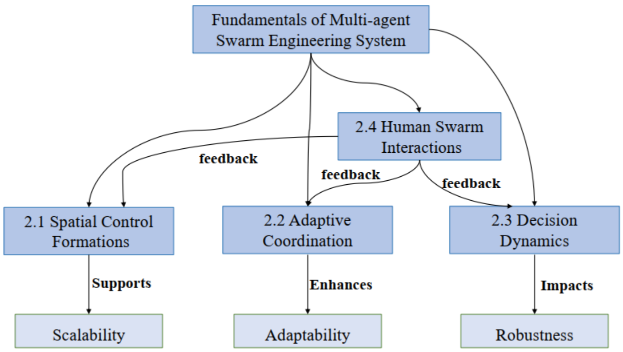

FIGURE 1. **Fundamentals of swarm engineering system (MASES).**

The contributions from various researchers, drawn from Google Scholar, provide a comprehensive summary of contributions from researchers, offering valuable insights into the fundamental principles of MASES, as shown in Table 1. It highlights advancements in addressing key design metrics such as decentralization, scalability, robustness, and adaptability. Tables 2 and 3 expand on these contributions, identifying key findings and future research agendas. In Fig. 1, we provide a conceptual overview of the MASES framework, illustrating how the core principles contribute to achieving critical performance metrics such as scalability, adaptability, and robustness. The figure emphasizes the interdependence of the following components:

1. Spatial Control Formation: Ensures effective agent organization as the system scales, supporting scalability.
2. Adaptive Swarm Coordination: Allows agents to dynamically respond to environmental changes, enhancing adaptability.
3. Decision-Making Dynamics: Strengthen adaptability through decentralized and collective decision-making processes.
4. Human-Swarm Collaboration: Enhances robustness by incorporating human oversight, enabling systems to handle uncertainty and disruptions effectively.

The framework demonstrates the need to balance these principles for optimal MASES performance. It provides a systematic perspective on MASES design and optimization, highlighting how these interconnected components influence overall system functionality. The interdependencies outlined in the figure underscore the importance of integrating these principles to address the key challenges of scalability, adaptability, and robustness [20]. In conclusion, this framework supports the practical deployment of MASES in diverse applications, such as robotics, distributed computing, and healthcare. By laying the foundation for understanding how MASES principles tackle critical design challenges, it facilitates the analysis of research contributions and drives innovation in the field.

Multi-agent swarm-engineering systems (MASES) achieve decentralized and adaptive performance through the seamless integration of individual agent behaviors and environmental dynamics. Interactions between agents and their environment are the foundation of collective intelligence, as agents continuously adjust their behaviors in response to environmental cues and interactions with neighboring agents [21]. These dynamic interactions determine system performance and influence the ability of MASES to adapt to changes, optimize outcomes, and achieve goals in complex real-world scenarios.

To better understand and design such systems, conceptual models have been developed to highlight the critical components and interactions driving swarm behavior. The Agent-Environment Interaction Conceptual Model, illustrated in Fig. 2, identifies key elements—agents, states, actions, and rewards—that are essential for decentralized and adaptive system operations. This model underscores the role of these components in optimizing overall system performance, aligning with core MASES principles such as spatial control formation, adaptive coordination, decision-making dynamics, and human-swarm collaboration. Together, these components contribute to achieving key outcomes like scalability, adaptability, and robustness.

A central focus of the model is environmental optimization, which involves fine-tuning external conditions to enhance agent performance, adaptability, and collaboration. This optimization is critical for enabling MASES to effectively address the complexities of dynamic real-world applications. By refining environmental factors, MASES can better respond to changes and unpredictable conditions, ensuring resilience, scalability, and adaptability while achieving diverse objectives. This integrated approach not only strengthens the system's ability to manage challenges but also ensures that MASES can operate effectively across a wide range of dynamic and uncertain scenarios, making it a powerful framework for addressing complex, real-world problems.

The conceptual model outlined in Fig. 2 provides a detailed framework for understanding the foundational principles of MASES. These principles—spatial control formation, adaptive coordination, decision-making dynamics, and human-swarm collaboration—are critical for designing and optimizing swarm behaviors [22]. Each principle plays a distinct role in enabling the decentralized, self-organizing nature of MASES while ensuring scalability and adaptability in dynamic environments.

1. Spatial Control Formation:

This principle establishes the structure and distribution of agents within a swarm. By organizing agents effectively, spatial control enables coordinated interactions and ensures that the system operates efficiently.

2. Adaptive Coordination:

Adaptive coordination allows agents to dynamically respond to environmental changes and evolving task demands. This responsiveness ensures the system's flexibility and its ability to function effectively under varying conditions.

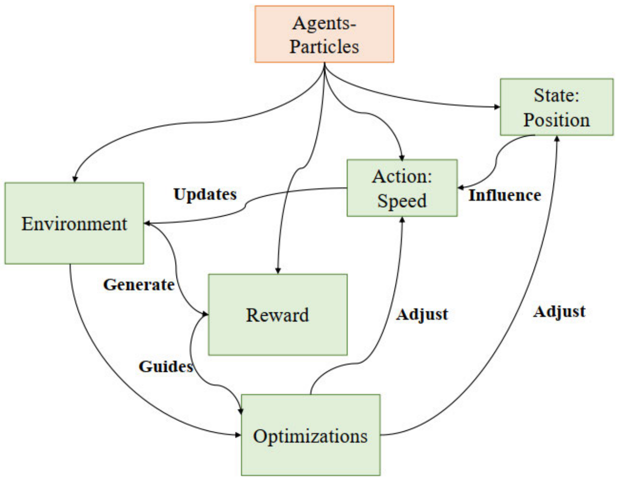

FIGURE 2. Agent-environment interaction model in MASES's [18].

3. Decision-Making Dynamics:

Decision-making dynamics govern how individual and collective decisions emerge from local interactions. By balancing decentralized decision-making with collective objectives, this principle supports cohesive and efficient swarm behavior.

4. Human-Swarm Collaboration:

Human-swarm collaboration ensures seamless integration of human oversight and intervention when necessary. This principle enhances system robustness and reliability by allowing humans to guide or adjust system behavior in response to unforeseen challenges. Together, these principles form the foundation for addressing complex challenges associated with MASES applications in robotics, healthcare, and distributed computing. By leveraging these principles, MASES can deliver decentralized, adaptive, and scalable solutions to meet the demands of real-world scenarios.

A. SPATIAL CONTROL FORMATION

Spatial control formation in MASES is the process by which agents self-organize into coordinated structures based on simple local rules to achieve global objectives. This process is guided by core principles such as decentralization, scalability, robustness, and adaptability, enabling agents to communicate with one another and interact with their environment, even under dynamic and uncertain conditions [23]. Decentralization allows agents to act based on local interactions without relying on centralized commands. This ensures that the system remains flexible and efficient, even when some agents underperform or fail. Moreover, decentralization inherently supports scalability, allowing MASES to expand or adapt seamlessly to meet varying levels of objectives and complexity [24]. A key feature of MASES, robustness ensures that the system is strong and resistant to disruptions, maintaining performance and stability in challenging conditions. Adaptability enables agents to modify their behaviors dynamically in response to changing environmental conditions, ensuring effective system operation in diverse and evolving scenarios [25].

Advanced methods, such as multi-agent reinforcement learning and real-time data processing, further improve MASES's coordination and flexibility. These techniques are particularly impactful in dynamic fields like disaster management and smart agriculture, where agents need to adjust their strategies rapidly to meet situational demands [26]. Mechanisms such as self-organization and self-assembly are crucial for enhancing the stability and efficiency of MASES. These processes enable agents to autonomously organize and construct functional structures, similar to how ants build bridges or walls for collaborative tasks [27].

Crucially, the integration of artificial intelligence (AI) and machine learning (ML) algorithms, such as reinforcement learning or deep neural networks, significantly enhances spatial control formation. These methods allow agents to learn optimal formation strategies and reconfigure spatial arrangements in real-time, directly improving the system's adaptability to dynamic environments and enhancing scalability by intelligently managing complex interactions. By employing these bio-inspired principles and advanced AI/ML techniques, spatial control formation in MASES provides a robust framework for decentralized, adaptive, and scalable solutions, enabling MASES to excel in managing complex, real-world applications across diverse domains.

B. ADAPTIVE COORDINATION

Adaptive coordination in multi-agent swarm-engineering systems (MASES) enables swarms to adjust their collective behavior in response to environmental feedback, mimicking natural organisms such as ants and birds. These adaptive behaviors are critical for activities like foraging, transporting objects, and group repositioning [28]. Decentralized communication facilitates these processes, allowing agents to operate autonomously without relying on centralized control [29]. The seamless integration of artificial intelligence (AI) and machine learning (ML) significantly enhances these capabilities by empowering MASES with real-time data analysis and learning. AI algorithms enable swarms to autonomously manage their strategies and effectively tackle emerging tasks in applications ranging from robotics to industrial automation [30]. Recent advancements in reinforcement learning have improved real-time coordination and decision-making, making MASES systems more efficient and scalable in dynamic applications such as search and rescue operations [31].

Nature provides numerous examples of adaptive coordination that inspire MASES design: Ants use pheromone trails to mark paths. For instance, ants utilize pheromone trails to mark efficient paths, while bees employ intricate waggle dances to guide swarm members to food sources [32]. MASES systems replicate these behaviors to coordinate complex tasks such as collaborative object transportation and precise positioning. For example, robots can maintain optimal spacing to ensure spatial synchronization and maximize overall team efficiency [33].

Ultimately, MASES systems are underpinned by four core principles that are crucial for enabling adaptive coordination: decentralization, scalability, robustness, and adaptability. Decentralization ensures that agents operate based on local interactions, eliminating the need for centralized control. Scalability allows MASES to efficiently adapt and expand as system requirements grow or tasks become more complex. Robustness makes MASES resilient to disruptions, maintaining functionality even in dynamic or uncertain conditions. Finally, adaptability empowers agents to adjust their behaviors in real-time based on environmental changes, ensuring continuous system performance. By thoroughly integrating these foundational principles with advanced AI and sophisticated adaptive coordination mechanisms, MASES systems gain the capacity to autonomously make decisions and adjust their actions in response to highly dynamic environments. This comprehensive capability facilitates superior self-organization and decentralized operations, offering innovative and resilient solutions to complex real-world challenges across domains such as robotics, disaster management, and industrial automation [34].

C. DECISION DYNAMICS

Multi-agent swarm-engineering systems (MASES) require robust decision-making capabilities to achieve the desired emergent swarm behaviors. Inspired by social animals like ants and bees, MASES rely on two primary forms of communication:

1. Direct Communication: Agents transmit information directly to other agents, enabling efficient and instantaneous coordination.

2. Indirect Communication: Agents use environmental markers, such as population density or information traces, to guide the actions of other agents and facilitate emergent behaviors.

These communication methods are critical for enhancing system performance as they support integrated decision-making processes. These communication methods are critical for enhancing system performance as they support integrated and decentralized decision-making processes. Recent advancements in artificial intelligence (AI) and machine learning (ML), particularly multi-agent decision-making algorithms, have fundamentally refined and advanced MASES's ability to make intelligent collective choices. For example, dynamic task allocation, often driven by ML algorithms, enables agents to adjust their responsibilities in real-time based on feedback from the environment and performance data, ensuring optimized system performance under varying and unpredictable conditions [35]. This includes leveraging advanced optimization techniques, often inspired by natural phenomena or AI, to enable agents to make collective decisions that maximize system efficiency, similar to how recent studies use bio-inspired optimizers or fuzzy logic for enhancing energy system performance [36].

MASES leverages this collective intelligence by seamlessly integrating and analyzing data from various sources to empower superior decision-making, allowing swarms to efficiently identify objects, assign tasks, and solve complex problems across diverse applications, such as optimizing workflows in industrial logistics systems where robots autonomously self-organize [37]. The deep integration of AI-driven algorithms and real-time feedback further enhances MASES's ability to dynamically adapt its decision-making processes and optimize collective outcomes.

MASES employs sophisticated coordination mechanisms such as

1. Continuous Interaction: Members maintain constant contact to make iterative adjustments.

2. Pulse Coupling: Members receive periodic signals to synchronize within larger, more complex swarms [38].

By combining these mechanisms, MASES achieves self-organization, self-optimization, and self-coordination in response to changing operational environments. Using locally informed decision-making, agents independently make decisions without relying on centralized authority, which promotes scalability. The deep integration of AI-driven algorithms and real-time feedback further enhances MASES's ability to dynamically adapt its decision-making processes and optimize collective outcomes. This continual improvement in decision-making and coordination empowers MASES to tackle a wide range of practical challenges, demonstrating its transformative potential in:

1. Robotics and Automation: Optimizing workflows and enhancing operational efficiency through intelligent task distribution and real-time adaptation.

2. Environmental Monitoring: Tracking and responding effectively to environmental changes by making informed decisions about data collection routes and anomaly detection.

3. Logistics: Streamlining supply chain processes with intelligent routing, inventory management, and autonomous material handling.

Advanced decision dynamics enable MASES to self-organize, adapt, and optimize their behaviors in real time. By leveraging decentralized communication, AI-driven algorithms, and real-time feedback, MASES can operate effectively across diverse and dynamic applications, demonstrating their potential to address complex real-world challenges.

D. HUMAN-SWARM COLLABORATION AND ETHICAL CONSIDERATIONS

Human-swarm collaboration is a vital component of multi-agent swarm engineering systems (MASES), particularly in applications where self-organizing swarm behavior benefits from human oversight. These interactions range from direct control, such as issuing commands, to indirect supervision, where MASES operate autonomously but remain capable of human intervention when needed [39]. This synergy is crucial for designing and deploying MASES effectively in practical scenarios.

Modern MASES leverage artificial intelligence (AI) interfaces to enable seamless human-swarm interaction. These interfaces combine graphical user interfaces (GUIs) with real-time swarm visualization and command input modalities, allowing operators to monitor swarm status and dynamically adjust operational parameters. Control schemes range from fully autonomous operation to adjustable autonomy, where the human operator can modulate the level of control, from supervisory monitoring to manual overrides, depending on mission criticality and environmental complexity. This flexible control framework ensures responsiveness, enables effective human oversight, and fosters trust in complex, dynamic environments such as healthcare, disaster response, and security [40].

Beyond operational interaction, the deployment of MASES necessitates a thorough examination of ethical considerations, which are inextricably linked to human oversight and public trust in autonomous systems [41]. Key ethical aspects include:

- Accountability: Establishing clear lines of responsibility for swarm actions and outcomes, particularly in unforeseen or failure scenarios.

- Transparency and Explainability (XAI): Designing systems whose decision-making processes can be understood and interpreted by human operators, fostering trust and enabling informed intervention.

- Bias Mitigation: Ensuring that AI algorithms governing swarm behavior do not perpetuate or amplify biases present in training data, leading to equitable and fair outcomes.

- Privacy and Data Security: Protecting sensitive information collected by swarm agents and ensuring the integrity of the system against malicious cyber threats.

- Societal Impact: Considering the broader implications of large-scale MASES deployment on employment, public safety, and human-technology relationships.

To operationalize effective human-swarm collaboration while addressing these ethical challenges, the framework incorporates several key features:

1. User Interface Design:

Human-swarm collaboration framework incorporates a multi-modal user interface designed for effective operator interaction. This interface provides real-time visualization of swarm positions, statuses, and task progress, enabling operators to maintain clear situational awareness. It also includes command and control panels that allow users to issue both high-level strategic goals and direct commands to the swarm [42]. Additionally, feedback mechanisms such as alerts for anomalous swarm behavior or potential ethical breaches notify operators when their attention or intervention is required. These components work together to reduce operator cognitive load and support efficient, ethically informed decision-making, which is crucial in high-stakes applications.

2. Levels of Human Control versus Autonomy:

The framework supports multiple levels of autonomy to suit different operational requirements. These modes include full autonomy, where the swarm operates independently with only periodic human monitoring; supervisory control, where the human operator provides strategic goals while the swarm executes tasks autonomously; and direct control, which allows operators to manually command specific agents or subgroups within the swarm [43]. This adaptability in control is vital for balancing efficiency with the need for human accountability and ethical governance.

3. Metrics for Measuring Interaction Effectiveness:

The framework employs several key metrics, like command response latency, which measures the time between an operator's input and the swarm's action. Operator workload and cognitive load are assessed using standardized tools such as the NASA Task Load Index (NASA-TLX) surveys. Task success rates compare system performance across different levels of human involvement, while user satisfaction, usability, and perceived trustworthiness are gauged through operator feedback studies [44]. These metrics provide valuable data that help refine user interfaces and autonomy settings, optimizing both efficiency and safety, ethical compliance, and overall trust in human-swarm interactions.

4. Safety and Oversight in Critical Domains:

For applications in healthcare, security, and other critical domains, the framework integrates robust safety and oversight mechanisms. Emergency stop commands and fail-safe behaviors allow operators to halt swarm activities immediately if necessary. Real-time anomaly detection systems continuously monitor swarm behavior and issue alerts to notify operators of potential problems [47]. These features help ensure compliance with regulatory standards and maintain reliable human supervision in mission-critical environments [45].

5. AI-Driven Autonomy:

AI empowers MASES agents to adapt and make decisions independently, ensuring continuous functionality even during local disruptions [46]. By integrating AI-driven autonomy with human oversight, MASES achieves a dynamic balance where human strategic thinking and decision-making are complemented by the swarm's decentralized, self-organizing capabilities. Furthermore, developing explainable AI (XAI) capabilities within MASES is paramount for fostering trust and enabling human operators to understand and audit the rationale behind autonomous decisions, particularly in ethically sensitive scenarios.

The synergy between human intelligence and swarm capabilities allows MASES to tackle complex industrial challenges with flexibility, adaptability, and efficiency. These systems can evolve and optimize in real time, addressing a wide range of issues in manufacturing, robotics, and automation. By integrating human decision-making with swarm efficiency, MASES provides innovative solutions that improve industrial processes, enhance operational efficiency, and adapt to dynamic conditions. This collaboration between human intelligence and swarm systems fosters a robust framework for dynamic responses to the growing demands of modern industries. By leveraging this synergy, organizations can improve adaptability, efficiency, and resilience in rapidly changing environments [47].

The foundational principles of multi-agent swarm-engineering systems (MASES) form the basis for designing decentralized, scalable, and adaptive systems capable of functioning in dynamic and uncertain environments. These principles ensure effective coordination and collaboration among agents, enabling swarms to achieve complex objectives without relying on centralized control. In Table 1, we highlight key research contributions that focus on the design and deployment of MASES, addressing essential swarm behaviors needed for decentralization, adaptability, and scalability.

The key features include

1. Emergence:

Emergence describes how complex collective behaviors arise from simple local interactions among agents. This principle underpins scalability, allowing MASES to grow and evolve with minimal centralized coordination.

2. Adaptive and Spatial Patterns:

Adaptive patterns emphasize the ability of agents to organize spatially and respond dynamically to environmental changes. These patterns directly address the need for adaptability by enabling MASES to optimize performance in real time and adjust to evolving conditions.

3. Self-Organization:

Self-organization enables decentralized system-level coordination, reducing the need for centralized control. This capability ensures that MASES maintains efficiency and scalability as the system expands in size and complexity.

4. Exploration and Localization:

Exploration and localization are critical for navigating uncertain environments and determining the spatial positions of agents. These capabilities enhance robustness, ensuring the system can operate effectively in dynamic and unknown settings.

5. Collaboration Mechanisms:

Mechanisms such as aggregate decision-making, task allocation, and synchronization ensure efficient collaboration and resource distribution among agents. These mechanisms allow MASES to reorganize and scale dynamically based on task demands, enhancing both scalability and adaptability.

6. Human-Swarm Interaction:

Human-swarm interaction combines human oversight and guidance with autonomous swarm behavior. This integration enhances robustness by enabling human intervention when necessary, ensuring system functionality under challenging conditions.

These key features provide MASES with the flexibility and resilience required to address real-world challenges across various domains. For example, robustness: Scalability: Emergent behaviors and collaboration mechanisms enable MASES to expand and adapt to increased system demands. Adaptability: Real-time optimization through self-organization and adaptive patterns allows MASES to respond effectively to changing conditions.

The fundamental principles of MASES create a robust framework for designing and deploying decentralized, adaptive, and scalable systems. By incorporating mechanisms for self-organization, exploration, collaboration, and resilience, MASES can address complex challenges in dynamic environments. These principles ensure that MASES is capable of evolving, optimizing, and maintaining functionality across a wide range of practical applications.

Each research contribution in Table 1 addresses a subset of the identified uncertainties and constraints. The "1" in each column indicates key elements that reflect the fundamental principles of swarm-engineering systems (MASES) and highlights the specific uncertainties and constraints each study focuses on. For instance, Zhang et al. [48] and Trianni et al. [49] focus on swarm robotics technologies, potentially addressing uncertainties such as environmental variability and interaction unpredictability. Their studies also deal with optimizing resource limitations or scalability issues. These studies target areas like swarm cognition, swarm coordination, routing optimization, and task allocation, each aimed at tackling specific challenges. Some studies emphasize environmental adaptability and self-organization, while others focus on decision-making and coordination.

In Table 1, we highlight the wide applicability of multi-agent swarm-engineering systems (MASES) across various domains while also describing their foundational principles, such as swarm robotics, optimization, and logistics. For example, in swarm robotics, MASES are applied to tasks like exploration, localization, and consensus-building, which enable autonomous systems to collaborate efficiently. In optimization, MASES focuses on task allocation, routing, and resource management in resource-constrained environments, with an emphasis on decentralized decision-making. Cognitive swarm systems further underscore the importance of self-organization and decentralized decision-making, where coordination and synchronization mechanisms ensure effective collaboration among agents in dynamic and uncertain conditions.

Additionally, we summarize the various uncertainties and limitations faced by researchers in these fields. These uncertainties include:

1. Environmental Variability: The unpredictability of environmental conditions.

2. Interaction Uncertainty: How agents interact with each other and with their environment.

3. Behavioral Complexity: The challenges of modeling and predicting agent behavior.

TABLE 1. Researchers' contributions illustrating the fundamental principles of MASES.

Paper	Emergence	Adaptive pattern	Spatial patterning	Self-organization	Exploration	Localization	Aggregated decision	Task Distribution	Collective Awareness	Synchronization	Swarm-human Interaction	Self-Healing	Self-Replication	Objective	Uncertainty	Constraints
Zhang et al. 2014	1			1					1					Swarm Robotics	Environment Variability	Resource Limitations
Garnier et al. 2007		1		1		1				1			1	Biological Inspiration	Replications	Scalability
Christian et al. 2008	1			1				1				1		Swarm Robotics	Behavior Complexity	Scalability Robustness
Shimming et al. 2008		1	1				1			1	1			Scalable Consensus	Agent Behavior	Decentralized Scalability
Duan et al. 2022		1			1		1	1				1	1	Adaptive Learning	Interaction Variability	Coordination Limitations
Trianni et al. 2011	1		1		1				1		1			Swarm Cognition	Interaction Uncertainty	Modeling Limitations
Deoca M.A.M. 2011		1			1		1			1				Cognition Integration	Feedback Uncertainty	Scalability Decision
Mullins et al. 2012			1		1		1		1			1	1	Decentralized Swarm	Environment Adaptability	Scalability
Chen et al. 2008	1			1		1						1	1	Consensus Protocols	Communication Disturbances	Communication Limits
Yasuda et al. 2013	1		1				1		1	1				Swarm Robotics	Communication Disturbances	Heterogeneity
Santos VG. et al. 2014	1	1						1	1	1				Formalized Swarm Models	Modeling Uncertainty	Real-time Limits
Junior L.S. et al. 2016	1		1	1			1				1			Swarm Optimization	Algorithm Transfer	Scalability
Junior L.S. et al. 2017	1			1		1					1			Swarm Intelligence Survey	Performance Variability	Algorithm Limits
Latifah et al. 2018	1	1		1						1	1	1		Swarm Coordination	Environmental Behavioral Uncertainty	Scalability
Sakthivelmurugan et al. 2021	1	1		1				1			1	1		Task Coordination	Data Communication Uncertainty	Scalability Adaptability
Peltokorpi V. 2008	1		1	1			1	1	1	1				Optimization Solutions	Resource Environmental Uncertainty	Complexity Adaptation
Rahmani et al. 2020	1	1		1		1					1			Routing Optimization	Solution Variability	Algorithm Efficiency
Oliveira et al. 2020	1	1	1	1	1	1	1	1	1	1	1			Logistics Optimization	Operational Variability	Scalability

4. Resource Constraints: Limitations in resources such as energy and processing power.

5. Communication Disruptions: Interruptions in the information exchange between agents.

6. Environmental Adaptability: The system's ability to adapt to changing conditions.

These factors directly influence the adaptability and robustness of MASES. The system must be able to respond to unpredictable changes while maintaining performance under uncertain conditions.

The constraints include resource limitations, which refer to the insufficiency of resources to meet all system requirements; scalability issues, which arise when maintaining system performance becomes challenging as the system grows; communication heterogeneity, referring to the variability in the communication technologies or protocols used between agents; and challenges related to unification and coordination due to the diversity of agents and the need for coordinated actions to enhance system functionality. These constraints can impact scalability (as performance may degrade with

TABLE 2. Key contributions to fundamentals of MASES: Insights from Table 1.

Category	Key Contributions	Representative Works
Coordination	Research on distributed coordination mechanisms, emphasizing multi-agent teamwork and scalable swarming behavior.	Coordination of multi-robot systems based on SI [50], Consensus on problems associated with scalable intelligent systems [51].
Communication Protocols	Required Strategies for efficient communication and navigation in dynamic environments.	Efficient Strategy for Collective Navigation Control in Swarm Robotics [52]. Wave Algorithm Applied to Collective Navigation of Robotic Swarms [53].
Energy Optimization	Techniques for minimizing energy consumption and optimizing swarm efficiency, focusing on incremental learning and resource allocation.	Incremental Social Learning in Swarm Intelligence Algorithms [54], Foraging Behavior Analysis of Swarm Robotics Systems [55].
Collective Decision-Making	Requirement of frameworks for collaborative decision-making inspired by natural systems.	Understanding Collective Decision-Making in Natural Swarms [56], Team Collaboration and Transactive Memory Systems [57].
Human-Swarm Interaction	Studies on Enhancements in Human-Swarm Interaction Using Bio-Inspired Cognitive Models.	Swarm Cognition: An Interdisciplinary Approach [58], The Swarm City - Characteristics of the SIs in the "Traditional Arab City [59].

an increase in system size) and adaptability (as the system must manage a variety of communication technologies and agents). Therefore, effective swarm systems must be designed to address these constraints and uncertainties, ensuring scalability and adaptability while maintaining the robustness necessary for reliable operation in dynamic and uncertain environments.

By addressing various combinations of these uncertainties and constraints, the contributions listed in Table 1 highlight different approaches for enhancing the functionality and resilience of MASES in fields like swarm robotics, optimization, and logistics. The table provides a structured and comprehensive overview of research contributions related to the core principles of MASES. Insights gained from these studies underline the pressing need for efficient coordination, synchronization, and adaptability in dynamic environments. Although uncertainties and constraints present significant challenges, overcoming them is vital for enhancing the functionality and resilience of MASES across various domains.

The insights presented in Table 1 serve as the foundation for the research contributions detailed in Table 2. In Table 1, we summarize key findings that highlight various uncertainties and constraints within multi-agent swarm-engineering systems (MASES), including the fundamental principles and critical behaviors that underpin these systems. These insights emphasize the urgent need for innovative solutions to address the challenges MASES faces in diverse fields, such as swarm robotics, optimization, and logistics. They also provide the groundwork for the more in-depth exploration presented in Table 2.

In Table 2, we explore in greater detail the key contributions driving the development of MASES, based on the insights from Table 1. These contributions are categorized into five main areas: coordination, communication protocols, energy optimization, collective decision-making, and human-swarm interaction. Each category highlights important advancements in addressing the core behaviors and design challenges of MASES. For instance, the coordination category focuses on distributed mechanisms that enable seamless multi-agent teamwork and scalable swarm behaviors. Research on multi-robot system coordination and the scalability of intelligent systems exemplifies this focus.

- In the communication protocols category, the emphasis is on strategies for effective communication and navigation in dynamic environments. This includes research on collective navigation control and wave-based communication technologies that aim to improve the responsiveness of the group under uncertain conditions.
- The energy optimization category explores technologies designed to minimize energy consumption and optimize resource allocation. Key research areas, such as incremental social learning and foraging behavior analysis, are pivotal in ensuring the efficiency of large-scale swarm systems.
- The collective decision-making category, inspired by natural systems, promotes frameworks for collaborative decision-making among agents. Studies on natural swarm behaviors and interactive memory systems demonstrate decentralized processes that enhance teamwork and resilience.
- Finally, the human-swarm interaction category focuses on enhancing collaboration between humans and swarms through biologically inspired cognitive models. Cross-disciplinary approaches to swarm cognition and insights into traditional social structures emphasize the importance of integrating human oversight into swarm systems.

The research findings presented in Table 2 offer a comprehensive framework that provides the foundation for the forward-looking research agenda outlined in Table 3. By addressing the key contributions in five categories—coordination, communication protocols, energy optimization, collective decision-making, and human-swarm interaction—Table 2 identifies important areas of progress that directly inform the future research necessary for the ongoing development of multi-agent swarm-engineering systems (MASES).

TABLE 3. Future research agenda for the fundamentals of MASES.

Research Agenda	Future Research Questions
Investigation of effective methods for coordinating multiple robots in dynamic and unpredictable environments that leverage scalability through techniques such as consensus algorithms and decentralized control mechanisms.	How can coordination protocols be enhanced to improve swarm performance in dynamic and unpredictable environments?
Exploring communication protocols for efficient collaboration in large-scale swarms, focusing on techniques like efficient information sharing (network flow models, data compression), decentralized communication (ad hoc networks, peer-to-peer), and swarm intelligence (wave algorithms, consensus protocols).	What communication protocols can be adopted to ensure efficient collaboration in large-scale robot swarms?
Developing energy-efficient robot behaviors to minimize power consumption using techniques like Energy-Aware Algorithms (power management), resource allocation (dynamic scheduling), behavioral optimization (swarm foraging), and learning algorithms (reinforcement and evolutionary).	What energy-efficient behaviors can be developed to minimize power consumption in swarm robots without compromising performance?
Developing frameworks to improve swarm decision-making efficiency, using collaborative decision-making (consensus protocols, opinion dynamics) and multi-agent techniques (negotiation models, voting systems, shared memory).	How can the accuracy and speed of collective decision-making be improved for complex swarm systems?
Investigation of human-swarm interaction with emphasis on usability and making systems more user-friendly, using bio-inspired cognitive models, human-robot interaction (gesture recognition, speech interfaces, and augmented reality), and swarm cognitive architecture.	How can user interfaces be designed to improve human control, monitoring, and collaboration with swarm systems?

In Table 3, we propose a forward-looking research agenda based on the insights and contributions presented in Tables 1 and 2. This agenda is designed to advance the foundational knowledge of MASES. In the area of coordination, the enhancement of coordination protocols aims to improve cluster performance in dynamic and unpredictable environments. This effort is a natural extension of the distributed coordination mechanisms discussed in Table 2, specifically targeting the scalability and environmental adaptability challenges highlighted in Table 1.

Furthermore, the communication protocols outlined in Table 3 focus on exploring more effective approaches to large-scale swarm collaboration. These initiatives address the communication interference and scalability issues mentioned in Table 1. Strategies such as network flow models and decentralized communication are intended to enhance the group's responsiveness and cooperation in complex environments. On the topic of energy optimization, the proposed research agenda emphasizes energy-efficient robotic behaviors. These strategies are closely integrated with the resource-constrained technologies discussed in Table 2, which are essential for ensuring the sustainable operation of swarm systems under limited resources, as highlighted in Table 1.

Additionally, the collective decision-making agenda aims to advance decentralized frameworks to improve decision-making efficiency. This agenda addresses challenges related to behavioral complexity and resource constraints while building on previous findings that demonstrate the practical benefits of improved decision-making in MASES. Finally, research on human-swarm interaction focuses on improving user interfaces to facilitate better collaboration between humans and swarm systems. By incorporating the insights from Table 2, this agenda aims to address the coordination and interaction challenges identified in Table 1, ultimately enhancing both user experience and system performance.

In this section, we provide a comprehensive exploration of the theoretical foundations of multi-agent swarm-engineering systems (MASES), with a focus on key concepts such as spatial control formation, adaptive coordination, decision dynamics, and human-swarm interaction. These elements are crucial for addressing core design metrics—particularly scalability, adaptability, and robustness—which are essential for the effective deployment of MASES across various applications.

In addition to the theoretical framework, three insightful tables summarize important information regarding the current state of MASES, enriching the discussion. A critical analysis of the research findings is presented as follows: Table 1 outlines the contributions of researchers to the fundamental principles of MASES, emphasizing key characteristics, objectives, constraints, and uncertainties; Table 2 summarizes important discoveries from prior research; and Table 3 highlights the forward-looking research directions necessary for advancing the field. Overall, these tables serve as valuable resources, providing researchers and practitioners with a clear understanding of the current challenges and innovations in the field. They underscore the urgent need to balance decentralization and scalability while addressing the inherent uncertainties and constraints of complex distributed systems.

Finally, the insights gained in this section advance the design of swarm-engineering systems by emphasizing the integration of collective intelligence with robust, scalable, and adaptive solutions. This integration is crucial for addressing real-world challenges in areas such as robotics and distributed computing, thereby paving the way for future research and innovation in MASES. Based on the contributions and future research directions outlined in Tables 1, 2, and 3, we now transition to the next stage: the design analysis of MASES. This phase focuses on the fundamental processes of system modeling and performance evaluation, which are critical for translating theoretical principles and research findings into practical, functional systems. In this design analysis phase, we will explore the key aspects of modeling swarm systems and methods for performance evaluation, laying the foundation for the development of more efficient and resilient MASES. With a strong grasp of these fundamental principles governing MASES behavior, the next crucial step involves translating these theoretical insights into practical implementations through systematic design analysis.

III. DESIGN ANALYSIS

The design of multi-agent swarm-engineering systems (MASES) is rooted in the principles of distributed, decentralized systems, where numerous simple agents collaborate to achieve complex goals without relying on centralized control. This approach is critical for ensuring that MASES operates efficiently across a wide range of practical applications. MASES design analysis uses various modeling techniques, including rule-based modeling, reinforcement learning, and performance analysis methods, such as agent-based models and integrated dynamic models. These methods are applied in the creation, analysis, and deployment of MASES to achieve core objectives such as decentralization, scalability, stability, and flexibility, especially in dynamic and complex environments [60].

By employing these techniques, design analysis contributes to the development of systems that not only address real-world problems effectively but also maintain high levels of flexibility. A key aspect of MASES modeling is capturing the behaviors of individual agents and their interactions within the system [61]. These models provide valuable insights into how local rules and agent behaviors combine to form advanced MASES, forming the foundation for effective system design. The discussion then shifts to performance evaluation methods, which are essential for benchmarking MASES and improving its efficiency. These methods evaluate whether the systems are scalable, adaptable, and robust, ensuring that they meet established performance standards.

The contributions of several researchers, drawn from Google Scholar and summarized in Table 7, offer key insights into MASES design analysis. This table highlights significant advancements in modeling approaches, performance evaluation, and system optimization. Building on these contributions in Table 8, we present key insights drawn from their findings, emphasizing progress and interdisciplinary approaches. Based on these insights, we present Table 9, which outlines future research directions in design analysis, focusing on developing innovative strategies to address persistent challenges such as decentralization, scalability, robustness, and adaptability. These trends aim to drive the evolution of MASES by integrating emerging technologies, refining existing methods, and expanding their applicability across diverse real-world domains. By focusing on these areas, researchers can enhance the effectiveness and impact of MASES in solving complex problems.

A. SYSTEM MODELING

Swarm system modeling is crucial for characterizing and enhancing the behavior of multi-agent swarm engineering systems (MASES) to improve efficiency and versatility across diverse environments [42]. Three primary modeling approaches have emerged to address the core design objectives of MASES: rule-based design, computational design, and reinforcement learning (RL).

Rule-based design involves agents following simple, local rules that enable them to respond to stimuli and accomplish tasks without centralized control [44]. This approach imbues the system with self-organization and self-healing capabilities, akin to natural swarms such as ants forming pheromone trails or birds maintaining flock formations. The decentralized interactions among agents allow the system to scale efficiently, maintaining robust performance even in unpredictable scenarios.

Computational design methods translate MASES challenges into formal optimization problems, employing algorithms like Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Genetic Algorithms (GA) [62]. These techniques enhance scalability by enabling agents to operate autonomously and efficiently at large scales. Moreover, computational design boosts adaptability by allowing MASES to dynamically adjust strategies in response to environmental feedback. This makes it particularly suitable for applications such as robot search-and-rescue missions or logistics, where agents must navigate obstacles or unevenly distributed loads [63]. Computational optimization, drawing from principles such as those seen in bio-inspired algorithms or AI-based methods like fuzzy logic [64], allows for the efficient exploration of solution spaces, enabling MASES to find optimal configurations and behaviors for improved performance.

Reinforcement learning (RL) offers significant advantages in MASES by enabling agents to iteratively refine their decision-making through trial and error, thereby improving their ability to adapt to dynamic, uncertain environments [65]. RL promotes scalability by training decentralized, coordinated agents and increases robustness by continuously incorporating feedback. This results in a system that performs reliably under uncertain conditions, such as

TABLE 4. Comparison of features across agent design approaches.

Feature	Rule-Based Design	Bio-Inspired Modeling (e.g., PSO, ACO)	Multi-Agent Reinforcement Learning (MARL)
Convergence Guarantees	Limited, dependent on rule design; stability is more about predictable behavior. It may not converge to optimal solutions in complex tasks [67].	Algorithm-specific; convergence to global optimum not always guaranteed; can be sensitive to parameter tuning [68].	Algorithm-specific theoretical guarantees are often weak in multi-agent settings; convergence is not always assured, especially in complex, non-stationary environments [69].
Sample Efficiency	Not applicable (rules are designed, not learned from data). Efficiency lies in the human effort to design effective rules [70].	Can be relatively sample-efficient for certain optimization tasks, especially in exploring large search spaces [71].	Typically low; requires significant interaction data (experience) to learn effective policies, making it challenging for real-world robotic systems [72].
Stability in Heterogeneous Environments	Can be brittle if rules are not designed to account for agent variations; requires explicit rule definition for different agent types [73].	Can be adaptable if the bio-inspired principles (e.g., local interactions) inherently handle heterogeneity; performance may vary [74].	Challenging due to non-stationarity and varying agent dynamics/reward functions; requires specific techniques (e.g., CTDE, robust reward design, transfer learning)[75].
Adaptability	Limited by the pre-defined rules; requires manual updates to handle unforeseen scenarios [76].	Generally good at exploring novel solutions within the problem space defined by the algorithm's principles [77].	High potential for adapting to novel situations through learning, but requires sufficient exploration during training [78].
Explainability	High; the behavior is directly determined by the defined rules [79].	Moderate; the emergent behavior can be complex to fully explain based on individual agent actions [80].	Low; the learned policies (especially with deep neural networks) are often closed boxes, making behavior difficult to interpret [81].
Computational Complexity	Generally lower during execution, but the design can be complex for large systems [82].	Varies depending on the algorithm (number of agents, iterations, interaction complexity) [83].	Can be high, especially during training and inference with complex policies and large state/action spaces [84].

exploration or search-and-rescue scenarios. Each modeling method carries distinct strengths: rule-based design is valued for its simplicity and modularity, computational methods excel in optimization and automation, and reinforcement learning is powerful for real-time learning and adaptive decision-making [66].

In Table 4, we provide a comparative summary of these modeling approaches across critical dimensions relevant to MASES design and deployment. The table highlights the inherent trade-offs: Rule-based systems offer high explainability and computational efficiency during execution, but can struggle with adaptability and stability in heterogeneous environments due to their reliance on fixed rules. Bio-inspired computational methods like PSO and ACO provide a moderate level of adaptability and sample efficiency in optimization tasks, though their convergence is often sensitive to algorithm parameters, and the resulting swarm behaviors may be difficult to interpret fully.

Multi-agent reinforcement learning (MARL) shows great promise for enabling complex adaptive behaviors within MASES. However, as Table 5 illustrates, MARL typically suffers from low sample efficiency, requiring extensive interaction data, and faces challenges related to convergence and stability, especially in dynamic, heterogeneous agent settings. Additionally, the policies learned via deep MARL can be difficult to interpret, raising concerns over explainability in critical applications. The computational demands for training and deployment of MARL agents can be substantial. Therefore, selecting a modeling strategy requires careful consideration of application-specific requirements, available computational resources, the need for explainability, and the operational environment.

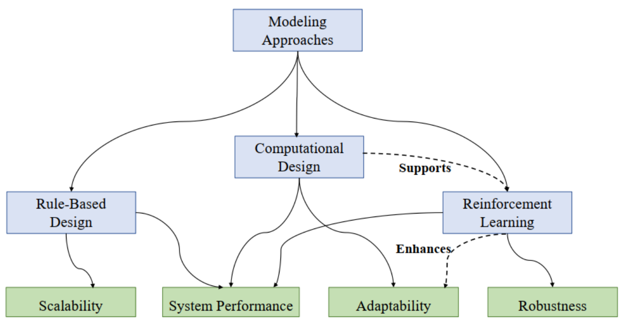

FIGURE 3. Swarm system modeling approaches.

Fig. 3 depicts the relationship between MASES design methods and their core objectives, emphasizing practical implications. Rule-based design underpins decentralized control, allowing the system to scale effectively by applying simple local rules to manage agent behavior, ensuring adaptability and fault tolerance. Computational design complements this by leveraging optimization techniques to simulate and refine configurations, yielding scalable and robust solutions that adapt to changing conditions. Reinforcement learning further enhances MASES by enabling agents to autonomously learn optimal strategies, improving adaptability and resilience against disruptions. Together, these methods form an integrated framework that equips MASES to address complex, large-scale challenges in dynamic, unpredictable environments.

MARL and bio-inspired modeling provide powerful mechanisms for fostering adaptive and robust swarm behaviors. However, their increased computational complexity can impact responsiveness and energy efficiency, especially in time-critical or resource-constrained scenarios such as disaster response or mobile robotics. To assess these trade-offs, we analyzed the computational overhead of the proposed frameworks by profiling processing time and memory usage and estimating energy consumption (using typical power models for embedded processors) under representative workloads simulating collaborative search-and-rescue tasks with swarm sizes ranging from 10 to 100 agents. The summarized results are presented in Table 5.

Our findings indicate that although more sophisticated algorithms improve coordination and reduce task completion time, they also increase processing delays, memory consumption, and energy usage. This heightened latency and resource demand may impair system responsiveness in real-time, energy-sensitive contexts. To address these challenges, our design employs a modular control architecture where lightweight rule-based controllers handle routine tasks, while complex learning algorithms are selectively engaged when greater adaptability is necessary. Additionally, approximate computations and heuristics reduce computational load during critical phases, and adaptive scheduling alongside energy-aware processing optimizes resource usage without compromising responsiveness. This balanced approach highlights the essential trade-off between algorithmic sophistication and practical constraints, ensuring that swarm engineering systems remain effective and deployable in real-world, time-sensitive, and energy-constrained environments.

B. PERFORMANCE EVALUATION

The performance evaluation of Multi-agent Swarm Engineering Systems (MASES) is essential for understanding how individual entities within a group act and collaborate to achieve coordinated behavior across various environments. Two primary evaluation methods—agent-based models and integrated dynamic models—are particularly effective for addressing issues related to scalability, adaptability, and robustness. However, it's also crucial to theoretically consider how practical resource limitations can impact the performance metrics assessed by these models, especially as the swarm size scales.

Agent-based models focus on the behavior and interactions of individual agents. They provide a micro-level view of how agents respond to local cues and self-organize to adapt to environmental changes [94]. These models play a crucial role in identifying potential scalability constraints by assessing whether agents can maintain efficiency in communication and coordination as the group size increases. For example, analyzing agent communication protocols and their bandwidth demands helps designers evaluate system performance and tackle the challenges of managing larger groups. The integration of AI-based algorithms further enhances flexibility, allowing agents to adjust their behavior based on real-time feedback. This adaptability improves system reliability, making it especially valuable in dynamic fields such as environmental monitoring and robotics, where conditions may change unexpectedly [95]. However, the effectiveness of these adaptive behaviors in large swarms can be limited by the available bandwidth for information exchange and the processing power of individual agents to execute complex algorithms in real-time.

On the other hand, integrated dynamic models provide a top-down view by focusing on the collective behavior of the group and assessing overall system performance. Tools like rate equations and system-level analysis help estimate task completion, resource consumption, and synchronization performance [96]. One key function of these models is to assess scalability, ensuring that the group remains efficient as the number of agents increases [97]. For instance, in robot swarms used for environmental monitoring, these models help determine whether performance improves with the deployment of more robots. By evaluating how communication load, coordination overhead, and task allocation efficiency change with the group size, designers can devise the optimal deployment strategy. This ensures that additional robots contribute to more efficient data collection and task execution without overloading the communication network or exceeding the processing capabilities of individual units. Moreover, integrated models analyze the group's ability to adapt to changes in the environment and resources, as well as the system's robustness in handling agent failures and environmental disruptions [98]. However, these models often operate under certain assumptions about resource availability, and their accuracy in predicting real-world scalability under constraints needs further consideration.

To theoretically illustrate the potential impact of such resource limitations on swarm performance with increasing agent numbers, Table 6 presents a simulated scenario where bandwidth and processing capabilities become constrained as the swarm size grows. In Table 6, we present a theoretical illustration of the potential impact of increasing agent numbers (N) on key performance metrics of a swarm system under simulated limitations in bandwidth and processing capabilities. The data suggests a clear trend of performance degradation as the swarm size grows and per-agent resources become scarcer.

Specifically, as the number of agents increases from 10 to 500:

- Simulated Bandwidth per Agent: Demonstrates a decreasing trend, halving approximately with each doubling of the agent count beyond 50, reflecting a scenario where a finite total bandwidth is increasingly shared among a larger number of agents.
- Effective Communication Rate: Shows a corresponding decline, indicating that despite the initially higher bandwidth per agent for smaller swarms, the actual rate of successful message exchange diminishes significantly as the swarm size increases. This suggests growing communication overhead, network contention, and increased latency.
- Average Task Completion Time: Exhibits a substantial increase, multiplying fivefold as the agent count rises

TABLE 5. Comparison of different agent design metrics.

Metric	Rule-Based Design	Bio-Inspired Modeling	Multi-Agent Reinforcement Learning (MARL)
Average Processing Time per Agent (ms)	10	25[86]	40[90]
Memory Usage per Agent (MB)	50	75[87]	120[91]
Estimated Energy Consumption per Agent (W)	1.2	2[88]	3.5[92]
Task Completion Time (s)	15	13	11
Responsiveness (Latency, ms)	50	80[89]	120[93]

TABLE 6. MASES performance metrics and scalability bottlenecks [99], [100], [101].

Number of Agents (N)	Simulated Bandwidth per Agent (kbps)	Effective Communication Rate (Messages/sec)	Average Task Completion Time	Processing Load per Agent (% of Capacity)	Coordination Efficiency (0-1, 1=Ideal)	Potential Scalability Bottleneck
10	100	80	5	15	0.95	-
50	100	75	6	30	0.9	Potential increase in communication latency
100	50	60	8	50	0.8	Increased communication latency, higher processing demand
200	25	40	12	75	0.65	Significant communication delays, approaching processing limits
500	10	15	25	90	0.4	Severe communication bottlenecks, high processing saturation, potential coordination breakdown

from 5 to 25 units. This indicates that the swarm becomes considerably less efficient at completing tasks as communication becomes slower, and coordination is hampered by resource limitations.

- Processing Load per Agent: Shows a marked increase, rising from 15 percent at 10 agents to 90 percent at 500 agents. This suggests that with larger swarms, individual agents are burdened with a greater share of the computational demands for coordination and task execution, potentially approaching their processing capacity limits and risking delays in real-time responses.
- Coordination Efficiency (0-1, 1=Ideal): Displays a consistent decrease, dropping from 0.95 for a small swarm to 0.40 for the largest simulated swarm. This signifies a significant reduction in the swarm's ability to effectively coordinate and work towards a common goal as resource constraints intensify with scale.
- Potential Scalability Bottleneck: Qualitatively identifies the primary limiting factor at different swarm sizes, transitioning from potential increases in communication latency in moderately sized swarms to severe communication bottlenecks and processing saturation in larger deployments, ultimately threatening coordination breakdown.

This theoretical illustration underscores the critical need to consider resource constraints when evaluating the scalability of swarm engineering systems. While a larger number of agents might intuitively suggest greater task-solving capacity, practical limitations in communication and computation can lead to diminishing returns and even performance degradation beyond a certain scale. The trends in this theoretical illustration suggest that despite the potential benefits of larger swarms, practical limitations in communication and computation can lead to significant performance degradation, including reduced communication effectiveness, increased task completion times, and potential coordination bottlenecks.

The design analysis of MASES focuses on optimizing system performance across various domains, ensuring scalability, adaptability, and robustness in practical applications. This comprehensive analysis provides valuable insights into the advancements in swarm system design and performance evaluation while also highlighting the challenges researchers face in ensuring effective operation within dynamic and uncertain environments. In Table 7, we present a detailed analysis of the key research contributions to enhance MASES modeling. This analysis emphasizes the essential design elements and performance metrics crucial for optimizing swarm systems, particularly for achieving scalability, adaptability, and robustness.

The table summarizes the key design elements identified by researchers, including cooperative behavior, behavioral rules, local interactions, agent behavior, and computational design. These elements are critical for optimizing agent behavior, task allocation, and resource management—all of which contribute to enhancing system performance and efficiency.

1. Cooperative behavior and local interactions ensure that agents can collaborate effectively, scale according to the system's needs, and adapt to changing conditions.

2. Behavioral rules provide real-time guidance for agent actions, allowing the system to operate under uncertainty and adapt to failures or disruptions, thus ensuring robustness.

Additionally, agent behavior and computational design are vital for ensuring scalability, enabling the system to increase in size and complexity without compromising efficiency. By integrating these design elements, researchers have deepened their understanding of self-organization, decentralized decision-making, and optimized agent interactions, crucial factors for building successful, scalable, adaptive, and robust swarm systems.

Furthermore, the table emphasizes the applicability of MASES across a variety of domains, including swarm robotics and autonomous systems. In swarm robotics, MASES allows drones and robots to collaborate and adapt to changing environmental conditions without human intervention, particularly in remote or unpredictable areas. In autonomous systems, MASES facilitates self-organization and cognitive decision-making, enhancing human-agent interaction, especially in situations with limited human supervision.

Additionally, we summarize several uncertainties and limitations addressed by researchers in MASES design analysis. These uncertainties include:

1. Environmental factors, where fluctuating conditions (such as weather or terrain) can impact the swarm's performance, particularly for drones or underwater robots.

2. Behavioral uncertainty arises from the unpredictability of agent interactions, which can disrupt coordination and convergence within the swarm.

3. Operational uncertainty refers to sudden and unforeseen changes in task objectives, requiring the system to adapt or reorient itself in real time. These uncertainties present significant challenges for designing systems that remain resilient and effective across varying conditions.

The constraints in MASES refer to limitations that may hinder the scalability and efficiency of these systems. These include computational constraints, where the processing power required to handle complex tasks and large-scale interactions is often insufficient, thus limiting the swarm's capabilities. Scalability challenges arise when the system becomes difficult to manage and maintain performance as the number of agents increases. Energy efficiency is another critical constraint, particularly in mobile systems like drones, where limited power resources must support prolonged operation. Additionally, as the swarm grows in size, design complexity becomes a significant factor, complicating the maintenance of system performance and coordination among an increasing number of agents.

Each research contribution in Table 7 addresses one or more of these uncertainties and constraints. The "1" in each column corresponds to the design analysis elements, uncertainties, and constraints that have been addressed by the respective researchers. These studies contribute to various aspects of MASES design analysis, such as cooperative behavior, agent interactions, and challenges related to scalability and energy efficiency. For example, Wu et al. [102] focus on cooperative behavior, agent actions, local interactions, and real-world uncertainties, with particular emphasis on resilient drone swarms and uncertainty testing. Similarly, Teodorovic [103] concentrates on swarm system validation, simulation accuracy, and interoperability limitations, with a focus on scalability and design complexity. By mapping the "1"s across the papers and topics, we can identify which studies address specific aspects of scalability, adaptability, and robustness in MASES design, offering valuable insights into the evolution of MASES research and its key focus areas.

By addressing different combinations of these uncertainties and constraints, the contributions in Table 7 highlight a variety of approaches to enhancing MASES modeling across multiple domains, including swarm robotics and autonomous systems. This diversity underscores the versatility and relevance of MASES in solving real-world challenges. The emphasis on cooperative behavior, behavioral rules, local interactions, agent actions, and computational design illustrates how these elements collectively enable decentralized decision-making, self-organization, and emergent swarm behaviors. Additionally, the discussion of uncertainties, such as environmental, behavioral, and operational uncertainties, reveals the complexities researchers must navigate when designing effective swarm systems. Identified constraints, including computational limits, scalability issues, energy efficiency, and design complexity, further underscore the need for innovative solutions to improve MASES performance. Overall, these findings offer valuable insights into the ongoing development of MASES research, guiding future studies aimed at overcoming these challenges and enhancing the implementation of swarm systems across various applications.

The results in Table 8 provide a solid foundation for justifying the importance of Table 9, highlighting key areas within the MASES design analysis that require further exploration. A comprehensive identification of essential design elements—such as cooperative behavior, behavioral rules, and computational design—underscores the multifaceted challenges that MASES must address. These elements are not only vital for optimizing swarm functionality but also emphasize the necessity of understanding how to tackle the various uncertainties and constraints that affect system performance.

In Table 8, we summarize the key features of MASES design analysis, drawing insights from Table 7 and organizing these contributions into six core focus areas.

1. Biomimetic modeling highlights the development of swarm intelligence (SI) algorithms inspired by the natural behaviors of ants, bees, and birds. These algorithms utilize genetic algorithms and neural networks to enhance the adaptability of swarms.

2. Decentralized control introduces autonomous decision-making technologies, enabling robots to function without relying on central control, which is crucial for safety-critical and multi-agent systems.

TABLE 7. Researchers' contribution for the design analysis of MASES.

Paper	Cooperative Behavior	Behavioral Rules	Local Interactions	Agent Behavior	Computational Design	Optimized Models	Real-World Uncertainty	Interaction Networks	Time to Convergence	Energy Efficiency	Coordination Efficiency	Modeling Complexity	Objectives	Uncertainty	Constraints
Wu.T.Z 2007	1		1		1		1		1				Resilient UAV	Testing Uncertainty	Evaluation Limitations
Yang.X.S. 2013		1			1			1					System Validation	Simulation Accuracy	Interoperability Limits
Dorigo et al. 2021	1		1	1				1					Adaptive Systems	Outcome Predictability	Replication Scalability
Teodorovic.D 2008	1	1		1	1								Robotics Trends	Solution Effectiveness	Integration Scalability
Yigit et al. 2019		1	1	1									Microrobotic	Interaction Variability	Communication Scalability
Singh.V. et al. 2016	1	1	1				1				1	1	Biomimetic Systems	Behavior Replication	Design Complexity
Tiwari.S.M. et al. 2018		1	1	1			1				1	1	Swarm Prediction	Environmental Sensitivity	Interaction Complexity
Selvaraj.S. et al. 2020	1	1		1	1			1			1		Swarm Optimization	Heterogeneity Impact	Design Challenges
Gautam.M. 2023	1			1	1								PSO Optimization	Environmental Variability	Adaptive Design
Kumoye.A.O. et al. 2020	1				1		1			1	1		Human-Swarm Trust	Trust Dynamics	Autonomy Variability
Bin.W. et al. 2001	1	1	1	1						1	1		Medical Robotics	Operational Uncertainty	Deployment Complexity
Reddy.E. et al. 2020	1	1			1	1							Swarm Interaction	Behavioral Uncertainty	Design Complexity
Winfield.A.F.T. et al. 2006	1		1	1			1				1	1	Reliability	Behavior Verification	Behavior Complexity
Rajarapollu.P. et al. 2020	1					1	1			1			UAV Resiliency	Disruption Impact	Design Gaps
Verdaguer.M et al. 2010	1	1	1			1		1			1		Underwater Swarms	Modeling Accuracy	Biological Model Application
Bonabeau.E. et al. 2010	1	1				1			1		1		Stability	Environmental Impact	Variability Constraints
Cuibus.M. et al. 2012	1					1	1						UAV Simulation	Environmental Variability	Simulation Interoperability
Oyelade.J. et al. 2019	1												Swarm Resilience	Failure Impact	Design Complexity
Figueiredo.E. et al. 2019	1	1			1		1			1		1	Mission Critical Swarms	Behavior Modeling	Real-World Applicability
Nair.S.C. et al. 2023	1				1	1				1		1	MBSE ROS Integration	Modeling Limitations	Translation Challenges
Chung et al. 2018							1			1		1	Aerial Swarm Autonomy	Environmental Interaction	Design Challenges

3. Adaptive learning explores algorithms that allow swarms to learn from real-time environmental feedback, with reinforcement learning enabling dynamic adaptation.

Regarding performance metrics, researchers have established benchmarks, such as clustering accuracy, energy efficiency, and decision-making speed, to evaluate the effectiveness of MASES in a range of applications. Algorithms designed to improve robustness and scalability ensure that swarms can operate effectively in challenging environments while scaling efficiently and maintaining resilience during failures. Finally, interdisciplinary collaboration promotes the integration of biological insights, artificial intelligence, and computational frameworks, which strengthens the adaptability and effectiveness of systems across fields such as robotics, urban planning, and cloud computing.

TABLE 8. Key contributions for design analysis of MASES: insights from Table 8.

Category	Key Contributions	Representative Works
Bio-Inspired Modeling	Development of SI algorithms inspired by natural behaviors like ants, bees, and birds, leveraging genetic algorithms and neural networks for adaptive swarm dynamics.	Research progress of SE [104], SI systems for transportation engineering [105], Swarm Intelligence for Multi-Objectives [106].
Decentralized Control	Techniques for decentralized decision-making and control, enabling independent robot operations without reliance on central controllers.	Towards the application of SI in safety critical systems [107], Integration of SI in MAS [108], Comparative SI analysis [109].
Adaptive Learning	Introduction of algorithms that allow swarms to learn from environmental feedback; exploration of dynamic models and reinforcement learning for real-time adaptation.	SI state-of-the-art computing [110], Adaptable SI framework [111], Automated Robot Communication System Using SI [112].
Performance Metrics	Establishing performance benchmarks such as clustering accuracy, robustness, energy efficiency, and decision-making speed across diverse applications.	Clustering algorithm based on SI [113], Enhancement of satellite images using SI [114], Critical Review on SI Algorithms [115].
Robustness and Scalability	Algorithms for robust operation under environmental challenges, ensuring scalability across large-scale systems and resilience against failures.	Data clustering algorithms [116], SI for clustering review [117], SI for detecting anomalies [118].
Interdisciplinary Collaboration	Encouraging integration of biological insights, AI, and computational frameworks to enhance system adaptability and effectiveness across domains such as robotics, cloud computing, and urban planning.	Survey of different SI algorithms [119], Control of SI and robotics [120], Swarm City Traditional Arab City Characteristics [121].

TABLE 9. Future research agenda for design analysis of MASES.

Research Agenda	Future Research Questions
Developing models inspired by animal behaviors (ants, bees, birds) for MASES design, focusing on scalability and adaptability. Using bio-inspired algorithms (e.g., ant colony optimization, particle swarm optimization) for tasks like drone path planning, ensuring effective scaling and adaptability in dynamic environments.	How can bio-inspired models contribute to addressing challenges in complex and dynamic environments?
Investigating decentralized control mechanisms for robustness and fault tolerance in MASES, focusing on consensus algorithms in multi-agent systems. These algorithms ensure that robots can operate autonomously without central coordination, handling unpredictable environments and scalable systems.	How can decentralized control strategies enhance robustness and fault tolerance in swarm systems?
Exploring adaptive learning approaches that integrate real-time algorithms to enhance swarm system performance and responsiveness. By incorporating reinforcement learning and dynamic models, these systems can adapt to environmental feedback, improving efficiency and responsiveness for unpredictable tasks.	How can adaptive learning algorithms be integrated to enhance system performance and responsiveness?
Investigating metrics to analyze the scalability, adaptability, and robustness of MASES, focusing on performance metrics and evaluation frameworks. Using clustering algorithms for tasks like satellite image analysis and energy efficiency metrics to evaluate the sustainability of robotic swarms in environmental monitoring.	What specific performance metrics should be developed to analyze the effectiveness of MASES accurately?
Investigate fault tolerance algorithms to ensure MASES remains efficient and adaptable as the swarm size increases. These algorithms address challenges in scalability, allowing systems to maintain robustness and operate smoothly in large-scale deployments and under resource constraints or environmental disruptions.	How can swarm systems scale effectively without compromising performance or robustness?
Exploring how integrating diverse fields can enhance the adaptability and effectiveness of MASES through interdisciplinary approaches, combining biology, AI, and computational modeling. This research focuses on cross-domain innovations to improve scalability and adaptability, leading to more robust MASES solutions.	How can interdisciplinary collaborations foster the development of innovative frameworks and methodologies in MASES?

These contributions are indispensable for advancing core MASES design metrics—scalability, adaptability, and energy efficiency—which are crucial for overcoming the uncertainties and constraints associated with MASES deployment.

The findings presented in Table 9 are essential in establishing the necessity of Table 8, as they provide a solid foundation for the integrated research agenda aimed at advancing MASES design and analysis. The insights derived from Table 9 highlight the importance of key areas, including biomimetic modeling, decentralized control, adaptive learning, robustness, scalability, and interdisciplinary collaboration. Each of these areas emphasizes innovative strategies to address the limitations and challenges identified in the previous table.

In Table 9, we propose a forward-looking research agenda for advancing MASES design and analysis, building upon the insights and contributions outlined in the earlier tables. This agenda begins by focusing on the potential of biomimetic algorithms, such as ant colony optimization and particle swarm optimization, to address scalability and adaptability challenges. This builds upon the discussion in Table 8, which underscores the importance of biomimetic modeling in dynamic environments and aims to optimize system performance, for instance, in applications like UAV path planning.

Regarding decentralized control, future research will center on improving fault tolerance and robustness through consensus algorithms, enabling MASES to operate autonomously in unpredictable environments without requiring centralized coordination. This is a direct extension of the decentralized control techniques discussed in Table 5 to enhance scalability and resilience within MASES systems.

The agenda also emphasizes the exploration of adaptive learning, particularly through reinforcement learning and dynamic models, which will enable MASES to learn from real-time environmental feedback and improve system performance. This will continue the exploration of adaptive learning methods highlighted in Table 5, focusing on enhancing efficiency and responsiveness. Additionally, the agenda calls for the development of performance metrics to analyze the scalability, adaptability, and robustness of MASES, particularly through clustering algorithms and energy efficiency metrics, addressing the challenges listed in Table 5. Research into fault-tolerant algorithms will be pursued to ensure MASES remains efficient and adaptable as the swarm size increases, building on the focus on robustness and scalability from Table 5.

Finally, the agenda underscores the need for interdisciplinary collaboration, combining insights from biology, artificial intelligence, and computational frameworks to create innovative approaches that enhance MASES performance across various application domains. This continues the emphasis on collaborative and interdisciplinary approaches found in Table 5, fostering further advancements in scalability and adaptability.

In this section, we conduct an in-depth analysis of the design approaches used in MASES, focusing on system modeling and performance evaluation strategies aimed at addressing core design metrics such as scalability, adaptability, and robustness. By examining various system modeling methods—including rule-based design, computational design, and reinforcement learning (RL)—as well as performance evaluation methods like agent-based models and integrated dynamic models, we emphasize their crucial role in transforming theoretical frameworks into practical implementations that effectively address the inherent complexities of decentralized systems.

The critical analysis of research findings is presented in three tables: Table 7 summarizes the contributions of researchers to MASES design analysis, highlighting key design features, goals, constraints, and uncertainties. Table 8 outlines significant findings from previous studies, while Table 9 identifies the forward-looking research directions required to advance the field. Together, these tables provide a comprehensive overview of MASES design analysis, emphasizing the importance of core design metrics in managing challenges related to coordination efficiency, fault tolerance, and system adaptability. They underscore the necessity of robust system modeling and performance evaluation methods, demonstrating their essential role in transforming theoretical frameworks into practical applications.

Furthermore, the successful implementation of these analytical frameworks in real-world applications highlights the importance of ongoing development processes and tools that support effective deployment. As we move into the next phase of this study, it is imperative to explore how frameworks and simulation environments can bridge the gap between established design principles and practical applications. This will ensure that MASES can operate efficiently and reliably across different scenarios and industries. This exploration will focus on the essential tools and simulations required to facilitate the successful application and operation of MASES, ultimately driving the further development of swarm engineering systems. To effectively realize these design objectives and evaluate their performance, robust development processes and sophisticated simulation tools are indispensable, as explored in the following section.

IV. DEVELOPMENT PROCESSES AND TOOLS

The development of multi-agent swarm engineering systems (MASES) requires a structured process, supported by various tools, to design, simulate, and test decentralized, adaptive, and scalable systems. This process spans from theoretical modeling to real-world implementation, utilizing these tools to model agent behavior, optimize system performance, and ensure practical applicability. The discussion on simulation suite frameworks introduces tools that facilitate early-stage concept modeling and theoretical exploration. Following this, insights into cognitive agent-based swarm engineering system frameworks are provided, detailing decision dynamics and adaptive behaviors by modeling the cognitive interactions between agents. An overview of simulation environment frameworks is also presented, emphasizing their role in establishing realistic testing conditions. These frameworks enable performance evaluations in dynamic and unpredictable scenarios. They allow researchers to test and refine multi-agent system designs before deployment, thereby enhancing decision-making, validating designs, and preparing the system for practical implementation.

Contributions from several researchers, summarized from Google Scholar, provide valuable insights into the development of simulation tools for enhancing MASES, as shown in Table 13. This comprehensive overview focuses on theoretical and practical advancements centered around core design metrics such as decentralization, scalability, robustness, and adaptability. Based on these contributions, key insights and future research agendas are outlined in Tables 14 and 15. By leveraging these frameworks and tools, the development of MASES can effectively address complex real-world challenges while ensuring system efficiency.

A. SIMULATION SUITE FRAMEWORK

Simulation suite frameworks offer high-fidelity, physics-based environments for modeling and testing swarm behaviors, enabling developers to simulate complex interactions and refine control algorithms effectively. In Table 10, we present insights into agent-based simulation frameworks, emphasizing the diversity of available options. Each framework caters to different needs, such as performance optimization, ease of use, and specific modeling requirements. These frameworks support various programming languages, including C#, Java, Julia, JavaScript, and C/C++, allowing users to select a language based on familiarity and project demands [121].

Additionally, many frameworks feature user-friendly graphical interfaces, such as CoSMoSim and DEVS-Suite, designed for users who prefer interactive design and visualization. For high-performance simulations, frameworks like FLAME and FLAME GPU are optimized for large-scale, compute-intensive tasks, leveraging high-performance computing (HPC) and GPU support. Frameworks like CoSMoSim and Evo Plex also provide modular and component-based approaches, enhancing customization and allowing users to tailor simulations to specific needs [122]. Moreover, frameworks like Gama, JADEX, and Agent Script focus on multi-agent systems, offering specialized tools for agent behavior, interaction, and coordination, which are essential in swarm engineering. The broad applicability of these frameworks in fields such as evolutionary theory, spatial simulation, and multi-agent systems underscores their versatility in addressing complex challenges.

Overall, these frameworks provide valuable insights, helping developers select the right platform based on specific goals, whether they prioritize ease of use, flexibility, performance, or managing the complexities of multi-agent interactions across various scientific and engineering applications.

B. COGNITIVE, AGENT-BASED FRAMEWORK

Cognitive agent-based frameworks enhance swarm engineering systems by simulating the decision-making processes, learning mechanisms, and social interactions among agents. These frameworks enable the creation of intelligent, autonomous agents that can learn, adapt, and make decisions based on feedback, thereby supporting the development of adaptive and robust systems [123]. In Table 11, we highlight several frameworks designed to model the cognitive, social, and emotional behaviors of agents. Each framework offers distinct advantages tailored to different user needs.

The cognitive modeling approaches across these frameworks vary. Some focus on individual decision-making, such as ACT-R and GOAL, while others integrate social learning and collective action, like SOSIEL [124]. Furthermore, these frameworks offer flexibility in programming languages, allowing developers to select the most appropriate framework based on their expertise or project requirements. Supported languages include Prolog, Java, and C#. Many frameworks, such as EBDI and ABC-EBDI, combine agent reasoning with cognitive processes. Others, like Jason and Soar, support complex multi-agent interactions and dynamic environments. Frameworks like SOSIEL incorporate social learning and memory systems, making them particularly well-suited for simulating swarm intelligence and social behavior. While ABC-EBDI and EBDI provide more accessible, no-code solutions, Jason and Soar offer greater flexibility but require more technical expertise [125]. In summary, these frameworks address a wide range of applications, from modeling individual agent behavior to conducting large-scale simulations. They equip users with the necessary tools to model and simulate various agent-based systems within cognitive, social, and emotional contexts.

C. SIMULATION'S ENVIRONMENT FRAMEWORKS

Simulation is a crucial phase in swarm engineering, allowing developers to design, test, and optimize agent behaviors before deploying systems in real-world scenarios. These frameworks provide realistic, dynamic virtual environments in which agents can interact, collaborate, and perform tasks, making them invaluable for testing scalability and real-world applicability [126]. In Table 12, we present insights into various simulation tools used for Reinforcement Learning (RL) and Multi-Agent Systems, each offering unique advantages tailored to different needs.

These simulation environments vary in programming languages, including Python, C++, and JavaScript, which provides flexibility for users to select a framework that best fits their expertise and project requirements [127]. The tools are highly customizable and scalable. For example, platforms like MAgent and MuJoCo support large-scale simulations, while others, such as Coach and DeepMind Lab, are tailored for real-world applications like robotics and autonomous driving. Moreover, these frameworks provide specialized support for various learning models, including those focused on swarm systems and agent interactions, allowing researchers to customize simulations according to their specific needs. Many platforms, such

TABLE 10. Agent-based simulation frameworks.

Sr. No.	Name of Agent	Language	Description
1	Actress Mass	C#	Used for multi-agent protocol and algorithm, algorithms were used for implementation of this agent framework, in which the agent's simulator framework was used.
2	Agents.jl	Julia	General-purpose, grid-based environments. The ABM framework is a Julia framework that provides data about structures and components and runs in batches to collect and visualize the data.
3	Agent Script	JavaScript	Based on Net Logo semantics, to provide the data about the agent-oriented programming model implementation used to promote the Script/JavaScript.
4	CoSMoSim	GUI-based	Component-based, modular, hierarchical modeling configures the input/output data for monitoring and visualizing in a unified modeling and simulation background.
5	DEVS-Suite	GUI-based	Rich visual modeling, component-based simulator, used for supporting the automatic design of experiments generating under super dense time data trajectories.
6	Evo plex	C++	In cooperative adaptive systems, an agent is represented under different scenarios, such as evolutionary graph theory, evolutionary crescendos, and game ideas.
7	Gama	Java GAML	The platform aims to provide developers with insight into building spatially explicit multi-agent simulations and is used for multi-agent simulation, modeling, and simulation environment development.
8	FLAME	C/C++	Based on agent-based applications and built on most of the existing systems. The ABM framework can spontaneously create different simulation plans, through which models can run efficiently on HPCs.
9	FLAME GPU	CUDA	GPU extensions to the FLAME framework used for different ABM building blocks are included, which helps the modelers focus on specific agent behavior and their simulation results.
10	Insight Maker	Runs & modeling browser UI	It provides a multi-method modeling solution framework to map out the theoretical models within a fluid and unified software environment.
11	Jaca MO	Agent Speak	A multi-agent programming framework provides three separate technologies: Jason-for programming autonomous agents, Cartago-for programming environment antiquities, and the last one is Moise (Mo), used for programming multi-agent groups.
12	JADEX	Java	Based on the agent motion with intellectual states. It provides the concept of a well-known acceptance, needs, and intent model at the design and execution stages.

TABLE 11. Cognitive, social, and affective agent frameworks.

Sr. No.	Name of Agent	Language	Description
1	ABC-EBDI	No code	Affective, BDI, agent's behavior, cognitive framework is modeled to consider the effects of position and orientation of nearby agents within the system, which also reflects the agent's conduct regarding the action to be performed.
2	ACT-R	ACT-R	Cognitive agents' behaviors provide the researcher with the framework to collect and measure the quantitative measures obtained from the participants within the simulation environment.
3	Cormas	Smalltalk	It provides the concept of anticipating different features, such as obstacles, within the environment. It is anticipated to assist in the design of ABM, along with observing and investigating simulation scenarios.
4	EBDI	No code	BDI merges with agent's abilities of cooperation theories with the agent's reasoning process; EBDI, by updating the BDI architecture, can merge several models that are based on the agent's reasoning process.
5	DALI	Prolog	An agent-oriented reasoning programming language offers the basic patterns identification and validation that are used for internal memory for modeling the agent-based system.
6	GOAL	GOAL	Programming language gives the building blocks for the execution of cognitive agents' tasks, in which agents drive their choice from their beliefs and goals, permit and accelerate the influence of an agent's ideas and goals, and configure its decision-making.
7	GROW Lab	Java	The framework that is designated to simplify the modeling, simulation, analysis & authentication of processes, which includes allocating the model with empirical realities of the MASES.
8	Jason	Agent Speak	Agent-oriented programming, "Agent Speak," provides for the development of agent systems based on significant abstract languages built on the BDI architecture.
9	Soar	Soar	Platform provides the development of systems that show intelligent agent behavior. It enables agents to communicate with the external environment and interface with agents through different languages.
10	SOSIEL	C#	Agent system modelling social learning, collective action, and agent cognition representation is generally done with cognitive architecture, which holds a memory module, learning module, and decision-making module.

as OpenAI Gym and Unity ML-Agents, incorporate state-of-the-art algorithms for training and evaluating intelligent agents [128]. In summary, these diverse simulation environments offer significant flexibility for experimentation, making them indispensable for researchers and developers working in reinforcement learning and multi-agent systems across various domains, from academia to industry.

TABLE 12. Simulation's environment frameworks.

Sr. No.	Name of Agent	Language	Description
1	Coach	Python	Presents the models for collaboration between an intelligent agent and several environments, displays a set of APIs used for experimenting with new RL algorithms, and allows the integration of new environments useful for autonomous driving & robotics applications.
2	Deep Mind Garage	TensorFlow, PyTorch	Framework for emerging and assessing RL algorithms in different circumstances, under which the simulations for training and testing have to be done. The toolkit offers a broad range of modular tools for executing RL algorithms.
3	Deep Mind Lab	Python	Provides a variety of challenging, customizable, 3D navigation plans for learning agents, which act as a test bed for AI-based research, particularly for DRL.
4	Deep Mind Sprite world	Python	It provides a 2D arena for modeling swarm systems with a variety of shapes, offers minor scaled experimentations with limited available computational resources, and constantly fluctuates in physical properties like position, size, color, etc.
5	GAZEBO	JavaScript	3D robot population simulation gives a high-quality graphics result and prebuilt robot models within the environments, recommends physics simulation execution with a higher degree of reliability, its sensors abilities, and provides interfaces for users and programmers.
6	MADP	C++	It provides mathematical modeling features for multi-agent decision-making modules like MMDPs, decentralized, partially observable MDPs, and partially observable stochastic games. It also offers the basic data used for modeling MADPs, such as the action-observation space.
7	MAgent	Python	Supports a huge number of agents, MARL, for supporting RL scenarios and addressing the scalability factors.
8	MuJoCo	C/C++	A physics engine is used for fast and accurate simulation, which provides better graphical and animated data for analysis and better solutions for improving swarm-based systems. It is proposed to develop a model-based optimization system.
9	Neural-MMO	Python	Provides a multi-agent simulation environment for efficient training and evaluation of agent interactions, by satisfying the agents' learning stability.
10	Open AI Gym	Python	provides the RL-based simulation environment to control tasks and permit custom agents to execute input parameters through training of RL algorithm-based models.
11	Open Spiel	C++/Python	Provides environments for cooperative and sequential movement of agents, to investigate learning dynamics and associated evaluation metrics.
12	PySC2s	Python	Deep Mind's Python component is used for the StarCraft II Learning Environment. PySC2 offers an interface for RL agents to interact with StarCraft 2, working based on observations and actions.
13	Unity ML Agents	Python	This toolkit presents implementations of the MARL system based on TensorFlow, which provides state-of-the-art algorithms to enable users to train intelligent agents.

D. THEORETICAL RATIONALE FOR SIMULATION-BASED VALIDATION

Although empirical validation through real-world deployment represents the gold standard for assessing system performance, practical constraints often limit the feasibility of large-scale physical testing during the early phases of SES research and development. As a result, this study emphasizes simulation-based validation, which, when rooted in well-defined theoretical models, offers a rigorous and scalable approach to evaluating swarm behavior [129].

The simulation frameworks employed in this work are designed to approximate real-world complexities by integrating stochastic elements, heterogeneous agent behavior, and dynamic environmental changes. These models are grounded in robust theoretical constructs from control theory, distributed AI, and bio-inspired computation, ensuring a high degree of relevance and transferability [130]. Moreover, the use of high-fidelity environments such as ROS and Gazebo, coupled with cognitive agent-based frameworks, enhances the realism of agent interactions and decision-making processes.

By systematically modeling decentralized control, fault tolerance, and adaptive coordination in varied scenarios, these simulations provide critical insights that inform future real-world testing. Thus, the reliance on simulation is not merely a substitute for empirical validation but a foundational stage that facilitates iterative design and theoretical refinement, laying the groundwork for subsequent physical deployment and field studies [131].

In Table 13, we provide a detailed analysis of key research contributions to enhance development frameworks for MASES. These contributions focus on essential features critical for the design and development of decentralized, adaptive, and scalable MASES.

These key features include:

1. Distributed Control: A system design where control is spread across multiple agents, eliminating reliance on a central controller and enabling decentralized decision-making and coordination.

2. Fault Detection and Recovery: The ability to identify and respond to system failures, ensuring that the system can recover quickly and continue functioning even in the event of faults.

3. Behavior Classification: A method of categorizing and analyzing the different behaviors of agents within the system to better understand and predict their actions and interactions.

4. State Representation: How the system's current conditions or status are modeled, helping agents understand their environment and make decisions accordingly.

5. Continuous Integration: The practice of continuously updating and integrating new features or improvements into the system, ensuring seamless operation and reducing the risk of system breakdowns.

6. Learning Adaptability: The system's ability to learn from experience and adjust its behavior to optimize performance in response to changing environments or conditions.

These features directly impact core design metrics, including scalability, adaptability, and robustness. For instance, distributed control and task decomposition allow systems to efficiently manage an increasing number of components without creating central bottlenecks, ensuring scalability. Fault detection and recovery mechanisms, along with error handling, enhance robustness, ensuring that MASES can maintain functionality and resilience even in the face of component failures or attacks. Learning adaptability and continuous integration promotes adaptability, enabling MASES to evolve and adjust in response to changing environments or system conditions. These contributions collectively advance the theoretical understanding of MASES and provide practical tools for creating scalable, adaptive, and resilient systems. Ultimately, these efforts bridge the gap between theoretical concepts and real-world applications, paving the way for future innovations in swarm engineering.

In this table, we also emphasize the applicability of MASES across various domains, including task flexibility, assembly accuracy, task reallocation, and task reliability. For example, in the realm of task flexibility, MASES can adapt to changing environments and objectives, enabling agents to take on different tasks as needed. This flexibility is particularly crucial in dynamic applications, where task demands change in real time, such as in search and rescue operations, environmental monitoring, or disaster relief.

Regarding assembly accuracy, MASES has proven especially useful in manufacturing and assembly processes, where multiple agents collaborate to perform precise assembly tasks. Swarm systems can leverage local communication and coordination strategies to work together, assembling complex devices or equipment with high precision. In terms of task allocation, MASES allows agents to dynamically redistribute tasks based on changing conditions, failures, or new priorities. This feature is essential for maintaining performance and efficiency when unforeseen events, such as agent failures or environmental changes, occur. Finally, with task reliability, MASES enhances overall reliability by ensuring that agents can collaborate to achieve objectives even when failures or disruptions occur. Through redundancy, fault tolerance, and self-organization, these systems allow agents to adapt, recover, and cooperate in real time, ensuring task success.

Additionally, we summarize several uncertainties and constraints identified by researchers that limit the performance and implementation of MASES. Uncertainties include the following:

Model Limitations: These refer to inaccuracies in the models used to represent real-world phenomena, which can lead to performance gaps or suboptimal behaviors in the system.

1. Convergence Challenges: It is difficult to achieve convergence to an optimal solution, particularly in high-dimensional or complex environments.

2. Environmental Sensitivity: Involves uncertainties arising from factors such as noise, sensor accuracy, and unpredictable changes in the operating environment.

3. Coordination Uncertainty: Focuses on the ability of the collective agents to effectively coordinate in real-time, especially when facing challenges in communication or environmental conditions.

In addition, constraints primarily include the following:

1. Hardware Limitations: Physical constraints such as sensor capabilities, computational power, and energy limitations, which may affect the system's effectiveness and performance.

2. Communication Limitations: Refers to factors such as limited bandwidth, interference, or sensor reliability, which restrict the effective exchange of information among agents.

3. Environmental Robustness: The system's robustness may be limited due to design and sensor constraints, particularly when facing complex or unpredictable environments.

4. Scalability Issues: As the size of the collective system increases, maintaining high performance becomes increasingly challenging due to computational limits and the complexity of coordinating a large number of agents.

These uncertainties and constraints present significant challenges for the design and application of MASES. Therefore, addressing these issues is crucial for enhancing the overall performance of collective systems.

In Table 13, the contributions of each researcher address various uncertainties and constraints. A "1" in each column indicates the specific key features, uncertainties, and constraints that the researcher addressed, such as task decomposition, fault recovery, and attack resilience. For example, Yun et al. [132] focused on optimizing obstacles and model limitations, while Calderon-Arce et al. [133] addressed scalability and logical optimization. Schlantz et al. [134] explored robot coordination and energy constraints. By mapping these "1"s across papers and topics, we can identify which studies focus on the scalability, adaptability, and robustness in the design and development of MASES. This mapping provides valuable insights into the evolution and key areas of focus in MASES research.

In summary, our analysis provides a comprehensive overview of the development of MASES, addressing critical challenges such as adaptability, scalability, and robustness. We have summarized the key features associated with effective performance in dynamic environments while also highlighting the significant uncertainties and constraints impacting MASES implementation. Advances in

TABLE 13. Researchers' contribution to the development process and tools used in MASES.

Paper	Distributed Control	Failure Detection & Recovery	Behavior Categorization	State Representation	Continuous Integration	Learning Capability	Task Decomposition	Environmental Awareness	Component-based Design	Error Handling	Resilience to Attack	Parameters Tuning	Resource Allocation	Objectives	Uncertainty	Constraints
TianYun.H. et al. 2008	1		1		1		1				1	1		Optimization Hurdles	Model Limitations	Solution Barriers
Calderon-Arce.C. et al. 2022	1	1		1				1		1		1		Logistical Optimization	Dynamic Constraints	Scalability Issues
Xin-She Yang. 2009	1	1		1	1		1		1			1		Design Trade-offs	Convergence Challenges	Dimensional Complexity
Schranz.M. et al. 2020		1	1				1				1	1		Robotic Coordination	Communication Constraints	Hardware Limitations
Foroutannia.A. 2021	1	1			1	1			1	1		1		Mission Reliability	Communication Constraints	Energy Limitations
T. Ray et al. 2001	1		1		1				1	1		1		Optimization Sensitivity	Local Optima Issues	Dynamic Adaptability
Bing Zhu et al. 2016	1	1			1	1			1		1		1	Mission Uncertainty	System Robustness	Scalability Issues
Pasechnikov.I. et al. 2021	1		1		1		1			1	1		1	Task Flexibility	Communication Reliability	Coordination Uncertainty
Anoop.A.S. et al. 2017	1		1	1		1			1		1	1		Assembly Precision	Material Limitations	Mechanism Complexity
Marco Dorigo et al. 2005	1	1	1		1			1		1	1			Collaboration Complexity	Calibration Challenges	Scalability Issues
S. Kazadi. 2005	1	1		1	1			1		1	1		1	Prediction Uncertainty	System Validation	Behavioral Complexity
Arvin.F. et al. 2018	1		1			1			1			1		Resource Optimization	Energy Constraints	Sensing Limitations
Cao.N. et al. 2019	1			1		1				1		1		Task Reallocation	Computational Burden	Dynamic Adaptability
Papoutsidakis et al. 2013	1	1					1	1		1				Resource Management	Hardware Constraints	Environmental Robustness
Xin-She Yang. 2014	1			1		1				1	1			Convergence Issues	Real-time Complexity	Algorithmic Uncertainty
Francesco Mondada. 2013		1		1			1	1			1			Reliability Challenges	Swarm Coordination	Scalability Issues
Efremov.M.A. et al. 2020	1	1			1	1			1		1			Environmental Variability	Network Scalability	System Flexibility
Kumar.Krishnan. 2005	1						1			1			1	Network Scalability	Transmission Efficiency	Swarm Coordination
Palaniappan. 2009	1						1			1	1		1	Network Reliability	Sensing Efficiency	Scalability Challenges
F. B. Bastani. et al. 2014	1			1	1	1			1					Movement Uncertainty	Task Complexity	Coordination Bottlenecks
C. M. Yang et al. 2006	1	1		1	1			1		1			1	Task Reliability	Performance Variability	Scalability Concerns
E. Sahin. 2007	1	1	1		1				1			1		Environment Sensitivity	Adaptation Challenges	Failure Resilience
H. Y. Wei. et al. 2015			1	1		1			1		1			Decision-Making Uncertainty	System Complexity	Real-time Coordination

coordination and communication have facilitated improvements in agent collaboration, while the integration of learning mechanisms has progressively enhanced decision-making capabilities. MASES demonstrates a broad range of applicability, including task flexibility, assembly precision, and task reliability. By mapping these contributions, we gain a deeper understanding of the ongoing evolution of MASES research, emphasizing the importance of scalability, coordination, and robustness while highlighting new challenges in optimizing MASES performance.

TABLE 14. Key contributions for the development process and tools: Insights from Table 13.

Category	Key Contributions	Representative Works
Integrated Frameworks	Development of frameworks that enable the seamless transition from concept to prototype and deployment in MASES.	Architecture analysis for swarm robot systems [136], Swarm robotic behaviors and applications [137].
High-Fidelity Platforms	Creation of simulation environments that replicate real-world conditions for testing and refinement of MASES.	Swarm Robotics Simulators and Platforms Review [138], SIN: A Programmable Platform for Swarm Robotics [139].
Simulation to Real-World Testing	Establishment of protocols and platforms for validating MASES performance in real-world applications after simulation.	Open-Source Experimental Setup for Real-World Implementation of Swarm Robotic Systems [128], Swarm Robotics Platforms Review [125].
Hybrid Algorithm Development	Development of hybrid models combining bio-inspired and computational approaches for enhanced swarm system performance.	Swarm intelligence-based algorithms: A critical analysis [140]. Hybrid-Digital-Mixed-Signal Computing Platform for Swarm Robotics [141].
Modular and Scalable Architecture	Design of adaptable and scalable architectures for MASES that support multiple applications and domains.	Architecture of Swarm Robotics System Software Infrastructure [142].

TABLE 15. Future research agenda for the development process and tools.

Research Agenda	Future Research Questions
Developing integrated frameworks to ensure seamless transitions between the concept, prototype, and deployment phases of MASES, which support flexible workflows, enabling real-time testing and refinement for enhancing the adaptability of MASES for diverse applications, from robotics to agriculture.	How can integrated design frameworks support flexibility and adaptability across diverse applications?
Developing high-fidelity simulation environments to test and refine MASES models before real-world deployment, ensuring accurate performance evaluation. Key simulation models, such as virtual environments, hardware-in-the-loop, and digital twins, are used for assessing scalability, reliability, and robustness in unpredictable environments.	What innovations in simulation technology can improve the accuracy and reliability of swarm performance evaluations?
Establishing protocols and platforms to validate MASES models in real-world settings after simulation testing to address challenges in transferring theoretical models to real-world applications, ensuring scalability and robustness during deployment. This includes testing frameworks, evaluation metrics, and platforms for post-simulation validation.	What best practices ensure the effective transfer of simulated MASES models to real-world applications?
Developing hybrid algorithms that combine bio-inspired models (swarm intelligence, genetic algorithms, ant colony optimization) with traditional computational techniques (machine learning, optimization methods) to enhance MASES performance.	How can hybrid algorithms balance computational complexity and efficacy to optimize MASES performance?
Developing modular and scalable architectures for flexible MASES designs. Key methods include modular design, distributed systems, and layered software architecture to enable effective scaling in multi-robot systems and diverse environments.	What design principles ensure the scalability, modularity, and adaptability of swarm systems for different scenarios?

The results in Table 13 highlight the importance of effectively addressing the challenges of adaptability, scalability, and robustness in MASES. By outlining the key features that enhance the resilience and functionality of MASES, we identify the critical uncertainties and constraints that must be overcome for successful implementation. These insights lead to Table 14, where we summarize the key contributions to MASES design and development based on the findings in Table 13.

In Table 14, we categorize these contributions into five key focus areas: integrated frameworks, high-fidelity platforms, real-world testing simulations, hybrid algorithm development, and modular and scalable architectures. Integrated frameworks emphasize the development of comprehensive structures that facilitate the transition from conceptual models to real-world deployment. High-fidelity platforms create simulated environments that replicate real-world conditions, addressing uncertainties related to system behavior in diverse scenarios. Real-world testing simulations extend this by providing protocols and platforms to validate MASES in practical applications, bridging the gap between theoretical models and real-world performance. Hybrid algorithm development combines biomimetic models with computational techniques to address challenges such as task allocation and system optimization. Finally, modular and scalable architectures focus on creating adaptable frameworks that directly address key challenges in scalability and adaptability for MASES deployment across various domains.

These contributions collectively provide a structured approach to improving MASES performance and addressing the uncertainties and constraints identified in Table 13. The key findings summarized in Table 14 highlight the direct contributions to the MASES design and development process, laying a solid foundation for advancing future research. Each of the identified key areas—integrated frameworks, high-fidelity platforms, real-world testing simulations, hybrid algorithm development, and modular and scalable architectures—provides the groundwork for overcoming critical challenges in MASES deployment. These contributions underscore the need for structured and adaptable processes to facilitate the transition from theoretical models to practical, real-world applications.

Building on these insights, Table 15 outlines a future research agenda for MASES development processes and tools. This agenda identifies the key research questions and areas that must be explored to further enhance MASES design and deployment, ensuring that the methods outlined in Table 11 are effectively leveraged. The agenda emphasizes the development of integrated frameworks that support flexible workflows across the concept, prototype, and deployment stages; the creation of high-fidelity simulation environments necessary for testing MASES models; the establishment of protocols for post-simulation validation of MASES models; the exploration of hybrid algorithms that balance complexity with performance; and the development of modular and scalable architectures to ensure adaptability across various MASES scenarios.

In summary, the research questions outlined in Table 14 stress the importance of continued progress in these areas to enhance MASES performance in a wide range of practical applications. Drawing from the findings in Table 13, the future research agenda is designed to address the pressing need for improvements in frameworks, simulation accuracy, real-world validation, and scalable architectures, thereby guiding the future development of MASES in real-world environments.

In this section, we present a comprehensive design analysis of Multi-agent Swarm Engineering Systems (MASES), focusing on various development tools and processes, such as simulation suite frameworks, cognitive agent-based frameworks, and simulation environment frameworks. These tools are specifically designed to address core design metrics, including scalability, adaptability, and robustness. Through this analysis, we highlight key contributions from researchers in addressing essential design metrics, including decentralization, scalability, resilience, and adaptability within MASES. These contributions underscore the importance of developing frameworks and tools while also considering the uncertainties and limitations that accompany MASES development.

Ultimately, the future of MASES lies in integrating advanced frameworks, simulation platforms, and hybrid algorithms designed to enhance performance and facilitate effective deployment across various domains. This integration is crucial for ensuring that MASES operates efficiently and reliably, paving the way for continued innovation in swarm engineering. Moving forward, researchers must keep refining these tools and frameworks, ensuring that MASES not only addresses current challenges but also adapts to the future demands of a constantly changing environment. The efficacy of these development tools is best demonstrated through their application in diverse real-world scenarios, which we examine in detail next.

V. APPLICATION OF MULTI-AGENT SWARM ENGINEERING SYSTEM

Multi-agent swarm engineering systems (MASES) have been applied across various fields, effectively leveraging decentralized coordination, AI-enhanced decision-making, and adaptability to address complex, large-scale challenges.

In industrial automation and logistics, swarm robotics optimizes warehouse management by collaboratively transporting heavy loads, scanning inventory, and restocking based on real-time demand. These applications often grapple with challenges like achieving seamless interoperability between different robot types and manufacturers, effectively fusing data from disparate onboard sensors (e.g., cameras, LiDAR, GPS) for accurate situational awareness, and integrating real-time operational data with existing control systems and databases [143]. Despite these complexities, the inherent scalability and decentralized intelligence of MASES, often augmented by reinforcement learning for optimized task allocation, are proving vital in significantly reducing costs, optimizing resource allocation, and improving efficiency, emphasizing intelligent resource management and seamless technological integration. In practice, such systems have been deployed in large-scale warehouses such as those of Amazon and Ocado, where they have demonstrated the potential to scale and enhance operational throughput [144].

In the agriculture sector, MASES has revolutionized practices by enabling precise crop health monitoring, automated irrigation, and targeted pest control. Swarms of drones actively monitor farmland, while ground robots precisely apply agricultural resources, minimizing environmental impact and boosting yields. However, this domain faces challenges such as integrating heterogeneous sensors from various drone and ground platforms, managing energy constraints, and ensuring reliable communication across large and complex environments. The MASES framework, leveraging principles of computational optimization and adaptive coordination—including insights from recent bio-inspired and AI-based optimization studies for energy systems [145]—addresses these challenges by supporting modular sensor fusion architectures that allow seamless integration of various sensor types and energy-aware communication strategies that ensure resilient operation, even under intermittent connectivity and limited power availability. Real-world trials in precision agriculture have shown that drones and robots working in tandem can reduce resource usage by up to 30 percent, providing clear evidence of the framework's applicability [146].

Environmental monitoring applications benefit from MASES by enabling scalable data collection in remote areas. Underwater swarms track coral health and water quality, while drone swarms monitor deforestation and animal migration. These systems provide continuous ecosystem surveillance, filling critical data gaps and integrating ecological health considerations into industrial practices. Notable deployments in monitoring marine ecosystems have successfully demonstrated that swarm systems can collect data over large areas with minimal human intervention, thereby enhancing environmental protection efforts [145].

In healthcare, MASES advances targeted medical procedures such as drug delivery, minimally invasive surgeries, and continuous health monitoring. Swarms of micro-robots deliver drugs directly to cancerous tissues and assist in achieving surgical precision, improving treatment outcomes while reducing side effects. However, critical challenges in this domain include stringent regulatory compliance, safety assurance, and effective human-swarm interaction. The MASES framework, particularly through its AI-driven autonomy and human-swarm collaboration components, integrates fail-safe swarm behaviors, explainable AI modules, and adaptive autonomy to meet stringent healthcare requirements while maintaining human oversight. For example, our system can implement automatic shutdowns or manual overrides during critical procedures, ensuring safety and compliance with regulatory standards, such as those outlined by the FDA [146]. Furthermore, ongoing research is exploring AI-driven decision-making that balances swarm autonomy with human control to improve patient outcomes.

Beyond these specific domains, general robotic swarm applications such as search and rescue, distributed exploration, and complex assembly inherently face unique challenges in coordinating heterogeneous agents, adapting to dynamic environments, and scaling effectively under diverse operational constraints. The MASES design, employing decentralized coordination algorithms and hierarchical control strategies, is critical to managing diverse robotic platforms and maintaining robust performance despite individual agent failures or rapidly changing mission parameters. Case studies in disaster response, for instance, have highlighted how swarm systems can adapt to highly dynamic environments, showing promise in both urban search-and-rescue missions and rapid logistics support during crises.

These domain-specific considerations enhance the practical relevance of swarm engineering systems by tailoring solutions to address sector-specific challenges while preserving their broad transformative potential. As MASES continues to evolve, its scalability, adaptability, and decentralized nature will drive innovations across diverse industrial and societal domains. Looking ahead, ongoing challenges include improving sensor accuracy in agriculture and ensuring robust regulatory compliance in healthcare, both of which are areas where further research and refinement are needed. By focusing on these evolving needs, we aim to further enhance the practical applicability and long-term impact of swarm systems in real-world applications.

VI. DISCUSSION

The design and deployment of multi-agent swarm engineering systems (MASES) present a compelling yet complex paradigm for revolutionizing industrial operations. This review has explored the dynamic interplay of theoretical and practical components that underpin this field. By examining theoretical frameworks emphasizing spatial control and adaptive coordination, we've highlighted the potential of MASES to enhance decision-making and enable dynamic responses to volatile industrial environments. The core design metrics of scalability, adaptability, and enhanced robustness, encompassing comprehensive fault tolerance and resilience against agent failures and environmental uncertainties, serve as crucial benchmarks for evaluating the efficacy of these systems in tackling contemporary industrial challenges.

Our critical analysis of design methodologies, encompassing rule-based computational design, reinforcement learning, and the increasing integration of bio-inspired and AI-based optimization techniques, reveals the diverse approaches employed to imbue MASES with intelligence and autonomy. Furthermore, the discussed performance evaluation methods provide essential tools for translating theoretical constructs into actionable frameworks, demonstrating the potential of MASES to improve coordination efficiency and significantly enhance fault tolerance and overall system resilience against unexpected agent failures and dynamic environmental uncertainties. The tabulated insights into researcher contributions serve as a valuable bridge, connecting theoretical advancements with the practical realities of implementation. This comprehensive integration underscores the fundamental need for MASES designs that can inherently adapt to uncertainty and bolster system resilience in real-world scenarios. The increasingly sophisticated integration of AI and machine learning into these methodologies further unlocks advanced capabilities for intelligent decision-making, adaptive learning, and scalable system growth, pushing the boundaries of MASES performance.

However, while the theoretical underpinnings and simulation-based evaluations of MASES offer significant promise, a persistent challenge remains in transitioning these advancements to robust physical deployments. As highlighted by the limited empirical validation in much of the reviewed work, the performance of MASES in uncontrolled, real-world environments often remains largely untested. This gap between simulation fidelity and deployment in physical environments necessitates a concerted effort in future research, particularly in overcoming practical hurdles such as ensuring robust system interoperability, developing advanced sensor fusion techniques for heterogeneous data, and streamlining the integration of diverse data sources in real-time.

Moving forward, it is crucial to prioritize empirical validation to rigorously assess the computational and theoretical models that underpin MASES. This includes designing and conducting real-world experiments that can evaluate the system's behavior under the inherent complexities and constraints of industrial settings. Addressing challenges such as scalability under practical resource limitations (bandwidth, power, processing), as theoretically explored in the Performance Evaluation section (Table 6), will be paramount. Future research should also focus on refining development tools and methodologies to facilitate the seamless translation from simulation to real-world application.

Ultimately, the transformative potential of MASES in industrial operations hinges on our ability to bridge the divide between theoretical promise and empirical validation. By focusing on rigorous real-world testing and addressing the practical challenges of deployment, the field can move towards realizing the vision of truly intelligent, autonomous, and resilient swarm engineering systems that can effectively address the complex demands of modern industries, while also rigorously addressing ethical considerations and fostering intuitive human-swarm interactions to ensure appropriate human oversight and public trust in these increasingly autonomous systems.

VII. CLOSING REMARKS

In conclusion, we present a compelling argument that multi-agent swarm engineering systems (MASES) have the potential to transform industrial operations by empowering industries with intelligent decision-making capabilities. By fostering innovation in system design and bridging the gap between theory and practice, MASES is poised to enhance operational efficiency and adaptability across various sectors. The insights emphasized in this paper highlight the importance of continued collaboration among researchers, industry leaders, and practitioners to refine these systems for real-world applications. As we move toward a future of increasing complexity, the integration of swarm engineering is critical for unlocking new dimensions of operational efficiency. Stakeholders must engage in this discourse, not only to understand the current landscape but also to contribute to the development of intelligent systems capable of navigating the intricacies of modern industry. This exploration aims to inspire further investigation into the synergies between MASES and industrial applications, ultimately paving the way for smarter and more resilient industrial ecosystems.

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